Data-Informed Disaggregation and Implicit Estimation of Emissions in Other Subsectors

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1. Introduction

Greenhouse gas (GHG) emissions inventories are crucial tools for setting mitigation targets, measuring progress toward goals and for accountability within the context of national and international commitments. There are several organizations – governments, academia and commercial entities – that are involved in data collection and preparation of inventories.

Of these, one of the most important sources of GHG data are official country inventories. All countries that are parties to the United Nations Framework Convention on Climate Change (UNFCCC) are required to prepare GHG inventories as part of their commitments toward Monitoring, Reporting and Verification (MRV) for which standardized requirements have been developed (UNFCCC, 2022). Annex 1 countries are required to provide annual inventories covering emissions and removals of direct GHGs – carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), perfluorocarbons (PFCs), hydrofluorocarbons (HFCs), sulfur hexafluoride (SF_6) and nitrogen trifluoride (NF_3) – from five sectors (energy; industrial processes and product use; agriculture; land use, land-use change and forestry (LULUCF); and waste), and for all years from the base year (or period) to two years before the inventory is due. Non-Annex 1 countries, on the other hand, have less stringent requirements and provide inventories as part of the national reports every 2-4 years and include other information such as mitigation actions, constraints and gaps, including support needed and received. As a result, these reporting requirements create a patchwork of GHG data that has varying levels of completeness, comprehensiveness, and detail.

Apart from official country inventories, several academic and research organizations also produce inventories at varying levels of granularity and recency. These inventories rely on statistics from the International Energy Agency (IEA; IEA 2022), the BP Statistical Review of World Energy (now produced by the Energy Institute; Energy Institute, 2024), the Food and Agriculture Organization's FAOSTAT database (Food and Agriculture Organization 2025), and the United Nations Statistical Division datasets (all of which rely on government data sources to some degree; UNSD/UNEP 2024). Examples of these inventories include the Emissions Database for Global Atmospheric Research (EDGAR; Crippa et al. 2021; Crippa et al. 2024; Crippa et al. 2025), PIK-Potsdam PRIMAP (Gütschow et al. 2021), and Global Carbon Project (Friedlingstein et al. 2022).

However, despite the existence of many global emissions inventories, several clear challenges exist for any policymaker seeking to access complete publicly available global greenhouse gas emissions estimates in practice.

The key challenges identified by Climate TRACE to accessing a complete global inventory were:

- **Geographic completeness:** Many otherwise high quality inventories informally referred to as "global" did not, upon close investigation, actually cover the entire globe. As a general rule, inventories were often relatively complete in wealthier countries, but frequently had significant gaps in the global South. EDGAR, for example, reports emissions for all sectors and countries, but sometimes has a time lag.
- **Sector completeness:** Nearly every inventory Climate TRACE was able to identify covered only certain sectors, rather than the total emissions for a geography. UNFCCC Annex 1 National Inventory Reports for example, report emissions for all sectors (but are not available in every geography and have a time lag).
- **Temporal completeness and recency:** Many existing inventories are not published every year. Most are also not published until at least 2 years after the emissions occurred. Some countries' most recent inventories–particularly the official UNFCCC inventories of non-Annex 1 countries–were as much as 10 years ago. A few exceptions exist, most notably Carbon Monitor which publishes emissions estimates only a week after the time period measured.

In addition, the lack of completeness and standardization of emissions inventories can make data harmonization and cross-validation difficult or, in some cases, impossible.

The goal of Climate TRACE is to address these challenges by providing a globally complete and comprehensive national level GHG inventory that is:

- Continually updated every year with emission values up to at least the preceding calendar year and the current year's emissions projected. In 2025, Climate TRACE GHG estimates were updated to include 2015-2025 national level emission estimates;
- Estimate emissions from all emitting subsectors with as much granularity as possible;
- Attributes all GHGs down to the emitting source/asset.

Table S3, in the supplemental section, provides a list of Climate TRACE sectors with IPCC equivalent sector coverage. Despite these continuing efforts, due to the nature of some subsectors (too many facilities, lack of existing data sources, lack of any visible satellite-derived signal), some emissions were still being estimated at the country-level only and aggregated at the sectoral level. For example, roughly 6.4% of 2015-2022 GHG emissions, or about 29 billion tonnes of carbon dioxide equivalent (CO_2e), were attributed to "*other-manufacturing*" in the 2023 (Version 3.0) release of Climate TRACE data, which are now broken down into more granular sectors. *Other-manufacturing* consisted of emissions from various industry and manufacturing subsectors, including from the production, creation and/or processing of electronics, machinery manufacturing

and other sectors estimated at the country-level despite each of these having a unique sectoral code in the United Nations Framework Convention on Climate Change (UNFCCC; UNFCCC, 2024). Furthermore, other existing inventories also combine these subsectors, making it difficult to quantify the climate impact of these individual industries. A deeper understanding of the contribution of each of these individual industries/sectors to GHG emissions will better inform decision making at the global, country and local levels.

In order to have a complete inventory of all GHG sources in every country/ region in the world, Climate TRACE members and contributors use several measurement and estimation techniques:

1) **Machine Learning + Satellite Measurements**: These models utilize detectable signals in satellite imagery to quantify activities at a facility and then subsequently estimate emissions based on the change in these activities using machine learning algorithms (e.g. steam plumes from power plants or heat from steel plants).
2) **Statistical Modeling + Satellite Measurements**: These models utilize satellite imagery to identify and characterize emission causing sources, such as the facility type, its size, and use statistical approaches to estimate emissions (e.g. cattle operations and wastewater treatment plants).
3) **Statistical Modeling + Reported Data**: These models utilize reported data gathered from public or proprietary sources and estimate emissions using statistical models (eg. aviation emissions using the ICAO methodology or oil refining emissions using the PRELIM model).
4) **Data-informed Emissions Disaggregation**: This method estimates emissions for facilities for which very little information is available (such as their location, capacity, type of manufacturing process, production quantities, etc). It utilizes existing GHG inventories and disaggregates sectoral emissions to all the known facilities in the country, which are obtained from public registries, industrial databases, and supply-chain (e.g. textile manufacturing and food and beverage manufacturing). Where individual facilities report their own emissions to a governmental registry, those reported values are used directly, and only the remaining unattributed portion of the sectoral total is disaggregated across the other known facilities.
5) **Implicit Estimation**: For all other sectors where facility level emission could not be estimated using any of the above 4 methods, emissions are ‘implicitly’ estimated using existing GHG inventories, subtracting out the subsector(s) that Climate TRACE estimates using methods 1-4.

Estimates from Climate TRACE members and contributors using techniques 1, 2, and 3 make up 81% of global emissions. This document describes techniques 4 and 5, where Climate TRACE uses the Intergovernmental Panel on Climate Change (IPCC) common reporting framework (IPCC 2006) along with other inventories to develop novel techniques to complement other GHG measurement methods in forming a global, comprehensive greenhouse gas inventory.

Here, we outline a process for the **Data-informed Emissions Disaggregation** technique, i.e. estimating both country- and asset-level emissions for several manufacturing subsectors originally encapsulated in *other-manufacturing* in previous Climate TRACE releases. The goal is to slim down and/or identify the unknown manufacturing subsectors lumped into *other-manufacturing*, common amongst GHG inventories. Multiple existing country-level inventories and economic datasets, as well as asset-level information from several data sources, were employed within a data-informed framework to disaggregate emissions to specific locations. This work is part of a larger effort across all Climate TRACE sectors to improve completeness of emissions at the asset and source levels, and also to increase both the granularity and accuracy of these emissions estimates. The following manufacturing subsectors were estimated using this method:

- *Glass*
- *Lime*
- *Textiles, leather, and apparel (at the asset level only)*
- *Food, beverage, and tobacco*

These subsectors together accounted for approximately 1.5% of global emissions in 2025.

For all other subsectors where Climate TRACE currently does not have a data-informed disaggregation estimation methodology yet due to the lack of facility-level information, emissions were implicitly estimated - technique 5 - using the Emissions Database for Global Atmospheric Research (EDGAR) and Food and Agriculture Organization of the United Nations (FAOSTAT) emissions inventories in order to provide a comprehensive emissions inventory at the country-level. The following sectors and subsectors incorporate EDGAR or FAOSTAT emissions data at the country-level:

- **Power sector:** *Other energy use*
- **Transportation sector**: *Railways; Other transportation*
- **Buildings sector:** *Other onsite fuel usage*
- **Fossil fuel operations sector:** *Other solid fuels; Other fossil fuel operations*
- **Manufacturing sector:** *Wood and wood products, Other metals, Other chemicals, Other mining and quarrying, Textiles, leather, and apparel (country-level), Other manufacturing*
- **Agriculture sector:** *Enteric fermentation - Other; Manure management - Other; Other agricultural soil emissions*
- **Waste sector:** *Biological treatment of solid waste; Incineration and open burning of waste*
- **Fluorinated gases sector:** *Fluorinated gases*

These subsectors together accounted for approximately 17% of global emissions in 2025.

The overall approach, followed by specific methods for quantifying country-level and asset-level emissions are presented in Section 2. Section 3 outlines the results from this analysis, while in Section 4, the implications and future considerations are discussed. Furthermore, starting in November 2025, Climate TRACE is now providing emission reduction solutions (ERSs) to each

sector to understand how sector mitigation strategies can be used to reduce emissions. ERS is discussed further in Section 5 below.

2. Materials and Methods

Here, we describe the two approaches, the **Data-informed Emissions Disaggregation** and **Implicit Estimation,** used to estimate emissions for the other subsectors. Emissions using these methods were estimated at the country- and asset- levels for years 2015 through 2025 at annual temporal resolution. Data were downscaled to monthly resolution using methods described in Raniga et al. (2024).

2.1 Data-informed Emissions Disaggregation

2.1.1 Datasets employed

To identify manufacturing facilities and to generate emissions estimates, various emissions inventories, asset location and emissions data, capacity and production data, and economic data were employed. This includes:

1. Country-level emissions quantities for each sector were taken directly or derived from the *Emissions Database for Global Atmospheric Research (EDGAR)* (EDGAR, 2025) and *Community Emissions Data System (CEDS)* (Hoesly and Smith, 2024). For the four disaggregated sectors - Glass, Lime, Textiles, leather, and apparel, and Food, beverage, and tobacco - the sector gap-fill equations are applied to the pre-publication ("staging") form of these inventories within the Climate TRACE data pipeline, before any facility-level data has been merged in (see Section 2.1.3.1).

2. Asset location, emissions, and other metadata were obtained from the following databases/locations.
    a. *European Pollutant Release and Transfer Register (EPRTR)* (European Environment Agency, 2023)*;*
    b. *Israel Pollutant Release and Transfer Register (Israel PRTR)* (Israel Environmental Licensing and Risks Prevention Division, 2024)*;*
    c. *US EPA Facility Level Information on Greenhouse gases Tool (FLIGHT)* (US EPA Office of Atmospheric Protection, 2024)*;*
    d. *Canada Greenhouse Gas Reporting Program (Can-GHGRP)* (Government of Canada, Environment and Climate Change Canada, 2024);
    e. *Open-Supply Hub* (2024), a curated registry of global float glass plants, a North American slaughter-facility registry, and other government websites.

f. Capacity data for lime facilities (including facilities in China) was obtained from LLM research; slaughter capacities (head per year) were obtained from the slaughter-facility registry and converted to carcass tonnage.
g. Country-level national establishment counts were obtained from the *Industrial Statistics Revision 4* (UNIDO, 2025), which was used as the completeness denominator (the expected number of facilities per country and sector).
h. Annual lime production data were obtained from the Lime Statistics and Information (USGS, 2025); These production statistics anchor the lime activity estimates.
i. Sector-specific emissions factors (EFs) and country-level production estimates were obtained from a partnership between ODE (https://ode.partners/en) and Climate TRACE, where LLMs were used to provide detailed breakdowns for emissions factors of different scopes and greenhouse gases. Each EF is compiled at the component level (combustion, process, and electricity) per product; the library used here retains only the Scope 1 (on-site combustion plus process) components (see Section 2.1.3.2.2).

Where multiple vintages of the same registry were available (datasets a through f above), all vintages were combined and de-duplicated so that the full reporting history of each facility (back to 2007 for E-PRTR) is retained.

2.1.2 Model development

In general, the workflow for generating emissions estimates for sectors within *other-manufacturing* starts with the existing EDGAR and CEDS emissions inventories, which provide country-level emissions estimates for each gas, by 'sector' (Figure 1). While both of these inventories aim for sectoral completeness, the partitioning of the emissions into sectors varies slightly. This allows us to piece together all emissions estimates from a given sector in a logical manner using a combination of these two inventories, in addition to the UNFCCC emissions database. The specific equations for deriving individual sectors are presented in Section 2.1.3.1.

The country-level total then serves as the starting point for the asset-level estimates. Facilities identified in public registries and industrial databases are placed into evidence tiers according to how much is known about each. Facilities with their own regulator-reported emissions keep those values; the remaining unattributed portion of the national total is disaggregated across the other known facilities in proportion to independent evidence of relative facility size. Finally, per-facility activity, capacity, emissions factors, non-CO_2 gases, confidence ratings, and uncertain estimates are derived, and the results are reconciled against the country total.

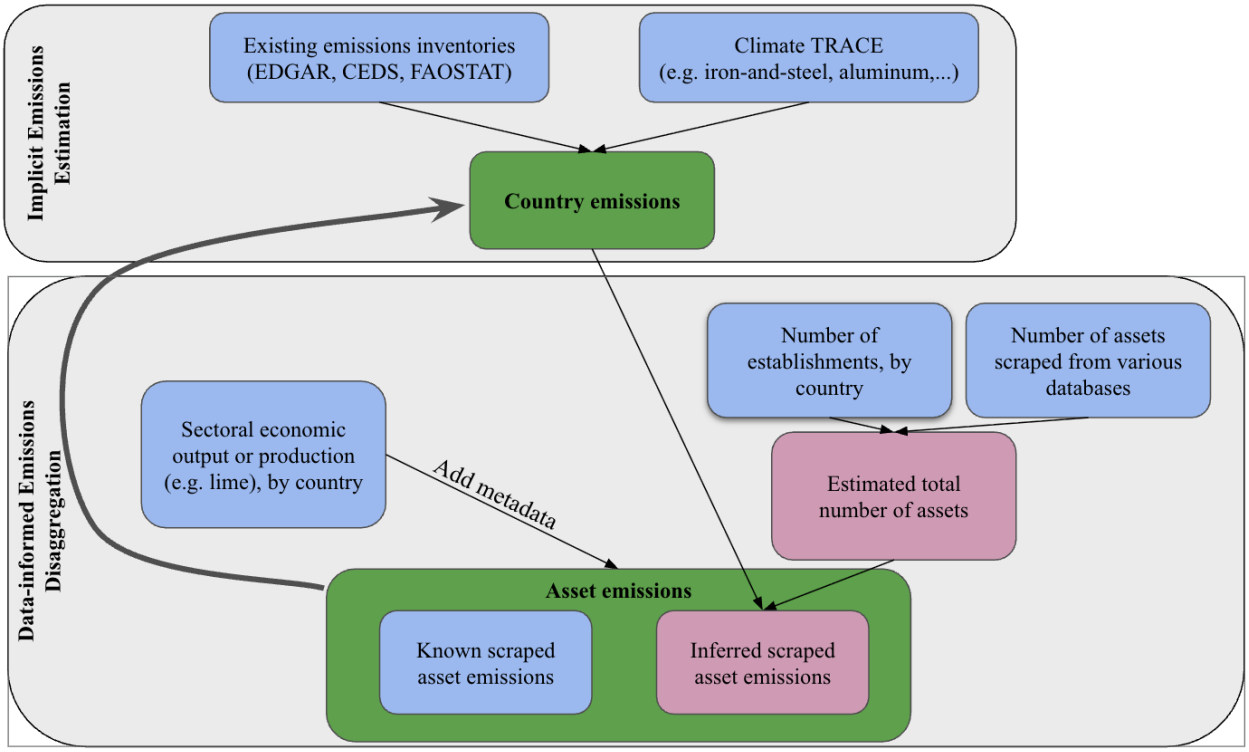


Figure 1. General process of creating country and asset-level emissions, along with associated metadata for each sector. The colors denote the type of information used in each step, with input data in blue, intermediate derived data in pink, and the final output data in green.

2.1.3 Methods

Country-level estimates exist for every sector, but only a subset contains information at the asset level. The method for estimating country-level emissions is described in Section 2.1.3.1, while the method for estimating asset-level emissions is described in Section 2.1.3.2.

2.1.3.1 Deriving country-level emissions

The following sections outline the equations used to quantify country-level emissions for the following sectors: *food, beverage, and tobacco*, *glass*, *lime*, and *textiles, leather, and apparel*. The CEDS or EDGAR sector naming convention is kept for clarity and other Climate TRACE sectors not described in this document but are used to derive the sectors discussed here are denoted with 'CT'. Country-level emissions from these two inventories are available through 2024 (EDGAR V2025) and 2023 (CEDS), respectively; years beyond the last available inventory year are forward filled from the most recent available year.

2.1.3.1.1 Food, beverage, and tobacco

In both EDGAR and CEDS, direct emissions from food, beverage, tobacco and wood manufacturing were categorized together, while combustion related emissions from food, beverage, and tobacco are reported separately from those from wood and wood products. Therefore we needed some way of estimating direct emissions from the subset of food, beverage, and tobacco manufacturing since Climate TRACE already accounts for wood and wood product manufacturing. The UNFCCC provides estimates of both direct (2.H.2) and indirect (1.A.2.e) emissions from food, beverage, and tobacco manufacturing for Annex I countries. Using these, we calculated a country-specific scaling factor as either the slope between 2.H.2 and 1.A.2.e of all annual available values 1990-2021, where a positive ($r > 0$) and statistically significant ($p < 0.1$) relationship was calculated based on linear regression, or the median ratio across all years for a given country. A p-value threshold of 0.1 was chosen to retain as many country-specific sector linear relationships as possible, rather than relying on a single global relationship that could misrepresent individual countries' sectors. We then applied this ratio to the CEDS 1.A.2.e emissions value for each Annex I country to estimate direct emissions from food, beverage and tobacco manufacturing.

As this relationship does not hold for non-Annex I countries, we assumed negligible direct emissions from food, beverage, and tobacco production in non-Annex I countries. Methane (CH_4) and nitrous oxide (N_2O) direct emissions were imputed by applying CO2-to-N2O and CO_2-to-CH_4 ratios that were calculated from facilities where both gases were reported.

The equation for estimating country-level food-beverage-tobacco CO_2 emissions was then defined as:

$$\textit{food-beverage-tobacco} = \textit{food-and-beverage-direct} + \textit{CEDS 1A2e_Ind-Comb-Food-tobacco}$$

where,

If non-Annex I countries:
$$\textit{food-and-beverage-direct} = 0$$

Else:
$$\textit{food-and-beverage-direct} = [[\textit{UNFCCC 2.H.2 / CEDS 1A2e_Ind-Comb-Food-tobacco}]] \times \textit{CEDS 1A2e_Ind-Comb-Food-tobacco}$$

Where, [[UNFCCC 2.H.2 / UNFCCC 1.A.2.e]] represents either the country-specific slope of the least squares line between the emissions from these two subsectors (where $r > 0$, $p < 0.1$), or the median ratio (else).

2.1.3.1.2 Glass

Glass manufacturing consists of both direct (2.A.3) and indirect (1.A.2.f) emissions. However, in CEDS, indirect emissions are aggregated together with indirect emissions from all non-metallic minerals (including cement, lime, and other carbonates). In order to disaggregate these indirect emissions, we estimated the contribution of glass combustion to CEDS 1.A.2.f by assuming that the ratio of direct to indirect emissions for each of these four mineral subsectors (i.e. glass, lime, cement, and other carbonates) were equal. With that assumption we calculated the relative contribution of glass direct emissions (2.A.3) to the sum of direct emissions from each individual subsector (i.e. 2.A.1 + 2.A.2 + 2.A.3 + 2.A.4), and multiplied that by the total indirect emissions (1.A.2.f). Therefore, the equation to estimate total glass manufacturing emissions is:

glass = EDGAR 2.A.3 Glass + glass-combustion

where,

glass-combustion = (CEDS 1A2f_Ind-Comb-Non-metalic-minerals) ×
EDGAR 2.A.3 Glass / (EDGAR 2.A.1 Cement + EDGAR 2.A.2 Lime +
EDGAR 2.A.3 Glass +
EDGAR 2.A.4 Other Process Uses of Carbonates)

2.1.3.1.3 Lime

Similarly to glass manufacturing, lime manufacturing consists of both direct and indirect emissions, the latter of which needs to also be disaggregated for CEDS data. Following the same formulation described above for glass manufacturing, the equation for calculating emissions from lime manufacturing is:

lime = EDGAR 2.A.2 Lime + lime-combustion

where,

lime-combustion = (CEDS 1A2f_Ind-Comb-Non-metalic-minerals) × EDGAR 2.A.2 Lime /
(EDGAR 2.A.1 Cement + EDGAR 2.A.2 Lime + EDGAR 2.A.3 Glass +
EDGAR 2.A.4 Other Process Uses of Carbonates)

2.1.3.1.4 Textiles, leather, and apparel

Here, CEDS textile and leather combustion-related emissions is set to "*textiles-leather-apparel*" which can contain, including but not limited to, linen, polyester, and silk.

textiles-leather-apparel = CEDS 1A2g_Ind-Comb-textile-leather

Note, at the country-level, this sector is filled using the “Implicit estimation” technique, but is included here, as its asset-level data are estimated using the “data-informed disaggregation” technique.

2.1.3.2 Asset-level data

Assets were obtained for the following sectors: *other-metals*, *other-chemicals*, *other-manufacturing*, *glass*, *lime*, *food-beverage-tobacco*, and *textiles-leather-apparel*. Emission quantities for each sector were obtained from their respective sources, where available. Due to a lack of production-level information, the metadata for *other-metals*, *other-chemicals*, and *other-manufacturing* were limited to: capacity = activity = 1 facility. Therefore, emission factors describe the emissions of a given gas, per facility. Emissions and emissions factors for assets with no available data were imputed using the default metamodeling methods, described in Moore et al. (2024).

For assets in the remaining sectors, *glass*, *lime*, *food-beverage-tobacco*, and *textiles-leather-apparel*, the following sections describe the **Data-informed Emissions Disaggregation** approach. The delivery covers 2015 through the most recent complete year, one row per facility per year.

2.1.3.2.1 Facility assembly

Registry union and identifiers. Available versions of each source registry are combined and de-duplicated, first on the publisher's own facility identifier and then on location, so that a facility's full reporting history is retained. Each facility carries one stable identifier across all of its years. Where a facility that Climate TRACE has already published moves to a new identifier in a refreshed registry pull, the previously published identifier is restored, so that the existing public asset is updated rather than duplicated.

Cross-publisher identity resolution. The same physical plant can be published by two different organizations (for example, a national emissions registry and a supply-chain database). Candidate duplicates are detected by co-location and by name-token containment (which captures the dominant real pattern, where one publisher qualifies the other's name with a site), with three protections: a single publisher's own distinct identifiers are never merged; names differing only in their digits (e.g. "Unit 1" / "Unit 2") are treated as distinct plants; and a blank name never corroborates a merge.

Sector audit. A publisher's sector tag is treated as a claim to be checked, not as ground truth. Several systematic corrections are applied: E-PRTR facilities filed under the ambiguous "production of cement or lime" activity code are resolved against Climate TRACE's own cement facility inventory (a facility within 750 m of a known cement plant is removed from the lime

population; the remainder are kept in the population but never contribute reported emissions, because their filings may include cement-process CO_2); livestock farms and under-cover crop operations carried under food-manufacturing codes are excluded, as they are covered by other Climate TRACE sectors; supply-chain product strings are routed to products on exact, validated mappings (for example, palm-oil mills are classified as oilseed processing rather than *grain* milling).

Facility lifetimes. Each facility carries an operating status per year. Sources observed in only a single year are treated as snapshots, where they are assumed to operate across the whole output window rather than to have "opened" in the observation year. A country's departure from a reporting regime (for example the United Kingdom leaving the E-PRTR) is not interpreted as facilities closing; those facilities are carried forward under the assumption of continued operation, and this is declared as a caveat. Explicit closure statements in source data override inferred statuses, and a terminal check guarantees that no facility-year believed to be closed ships with a quantity.

Locations. Redacted or placeholder coordinates (e.g. registry confidentiality) are withheld. Anything resolvable to the GADM 1 or GADM 2 level are published with their location as the centroid of the associated GADM 1 or GADM 2 boundary.

2.1.3.2.2 Emissions Factors

Emissions factors (EFs) are maintained in a single library, one factor per product (e.g. quicklime vs. hydrated lime; float glass vs. container glass; brewing vs. slaughter; weaving vs. garment assembly), compiled from LLM research and published industry sources at the component level: combustion and process components are summed into a direct (Scope 1) factor. Where no defensible EF exists for a product, none is invented; the global median factor across all products for that sector is employed.

Several physical-plausibility constraints are enforced on the factors and on the quantities they imply:

- Stoichiometric ceilings. Process CO_2 from mineral chemistry is bound by molar mass. For example, calcination of calcium carbonate cannot emit more process CO_2 per tonne of quicklime than the chemistry allows. For lime the process component of each facility's CO_2 is pinned at the library's process factor times activity, with the combustion component absorbing the remainder, so that no delivered row implies a process factor above its stoichiometric ceiling.
- Trust ordering. For glass, lime, and food, the trust order is emissions > activity > emissions factor, meaning reported emissions and independent production statistics are treated as truth, and the emissions factor absorbs any residual inconsistency and is flagged where it departs from the library value. For textiles, where production statistics are weakest (and

there is double counting of textile production mass because of the nature of the supply chain), the order is emissions > emissions factor > activity.
- EF review. Emissions factors failing physical review are replaced with the emissions factor of a similar product.

2.1.3.2.3 Evidence tiers

Every facility-year is placed into an evidence tier that determines how its emissions are estimated and how they are graded for confidence:

- Tier A: the facility's own CO_2, reported to a regulator for that year. Reported values are never modified.
- Tier Ac: the facility's own reported CO_2, carried forward or backward from its nearest reported year. A carried value is accepted only if it does not exceed 25% of the country-sector-year total (a carried filing is a statement about this plant, but not about this year).
- Tier A′ (Tier a 'prime'): CO_2 inferred from the facility's own reported non-CO_2 emissions via national gas ratios, subject to the same 25% envelope gate; an implied CO_2 above the gate is rejected rather than flagged, because an inference (unlike a reported value) is not evidence.
- Tier D+: no facility-reported emissions; the facility's share of the national residual is estimated from at least one facility-specific size signal or a same-product peer benchmark.
- Tier D0: no distinguishing evidence at all; the facility receives an equal share of what remains, and is graded at the lowest confidence.

For facilities reporting in some years but not others, the previous release's blanket backward/forward filling of all metadata is replaced by the gated Tier Ac carry above, which applies to the facility's own reported CO_2 only.

2.1.3.2.4 Allocation of the residual

Per country, sector, and year, the residual is the national total minus the sum of all anchored (Tier A/Ac/A′) facilities:

$$residual_{c,s,y} = \max\left(national\,total_{c,s,y} - \sum_{i \in \text{anchored}(c,s,y)} Q_i,\ 0\right)$$

Where, *anchored(c,s,y)* is the set of facilities in country *c* and sector *s* whose year-*y* emissions (Q) of a given gas is established by Tier A, Ac, or A′ (Section 2.1.3.2.3), and *anchored* Q_i is that facility's emissions of a gas. The residual is floored at zero; where reported facility values alone

meet or exceed the national total, no residual remains to allocate and the exceedance is accepted rather than corrected (Section 2.1.3.2.7).

Nonzero residual emissions are disaggregated across the remaining known facilities in proportion to a per-facility magnitude claim, in tonnes, according to three distinct pathways:

1. If an asset-level reporting entity with a minimum reporting threshold (e.g. EPRTR) exists for the country/sector, the unallocated asset is assumed to be below the reporting threshold, so a rank curve is fit to the anchored assets of the form:

   $$Q_i(r) = scale \times r^{\wedge}(-\beta)$$

   Where Q is the emissions of the rth ranked asset. Parameters *scale* and β are fitted using ordinary least squares on log Q against log r using the 10th to 90th percentile assets to prevent the extremes dominating the fit. The non-anchored assets, which are always ranked below the anchored assets, are ranked based on any available metadata: headcount, capacity, etc. Those without any relevant metadata are given the mean of emissions for the remaining ranks.

   After all asset emissions have been allocated, if the sum of asset emissions exceeds the country total, the country value is adjusted upward.

2. If no such reporting entity exists, there is no relevant reporting threshold to apply the first approach. Therefore, we apply a benchmarking approach that considers the same relevant metadata as the first approach that allows us to rank assets. We produce a first-pass estimate of emissions, Q' (Q prime), as:

   $$Q'_i = b_{s,p} \times m_i$$

   Where the peer benchmark, b, is the 33rd percentile of anchored facilities' emissions for that sector, s, and product, p, resolved down a four-rung fallback (same product and country → same sector and country → same product globally → same sector globally; the percentile, rather than a mean, both corrects for the reporting-threshold selection effect and prevents a single anomalous filing from setting the scale), and the size multiplier, m, comes from the best available facility-specific signal: publisher-stated capacity where it exists; otherwise worker headcount, compressed with an exponent of 0.6 because headcount is reported for whole sites in ordinal buckets; otherwise 1. Finally, normalization is applied as this pathway does not allow for upward adjustment of country values:

$$T = \min(\Sigma\, Q'_i, R)$$

$$Q_i = T \times Q'_i / \Sigma\, Q'_i$$

3. Without any distinguishing metadata, the residual is allocated equally to each asset as default.

Across the four affected sectors, 18% of residual emissions are allocated using the first approach, 79% using the second, and the remaining 3% default to the equal scaling. This is a significant improvement relative to V5.10.0 and earlier, during which the majority of non-anchored emissions were calculated using an equal distribution.

Additional constraints on the allocation:

- *Censored zeros are upper bounds*. A zero in the E-PRTR means "below the reporting threshold" (100 kt CO_2/yr under Regulation (EC) No 166/2006), which is a published upper bound, not a value. Allocated values for such facilities are capped at the threshold after the split, with the excess redistributed to the group's uncapped facilities.
- *Production ceiling*. A cohort of allocated facilities may not, in aggregate, claim more emissions than the country's production statistics could support (national production times the upper bound of the sector's EF band). Where reported facility level emissions alone already meet or exceed the ceiling, the contradiction suspends the ceiling for that country-sector.
- *No leveling*. Size evidence orders facilities but is never used to force them toward equality; where no evidence distinguishes facilities, the equal-share floor (Tier D0) is applied.

2.1.3.2.5 Activity, capacity, and capacity factor

Activity is expressed in tonnes of product per year in all four sectors. Where a facility's activity cannot be anchored to a production statistic, it is derived as emissions divided by the EF. For lime, national production statistics from USGS anchor the activity totals.

Capacity is defined as the facility's own peak activity across its delivered time series, unless a publisher-stated capacity exists and is consistent with the allocated activity. Under this definition the capacity factor (activity divided by capacity) equals 1 in the facility's peak year and is below 1 in all other years.

2.1.3.2.6 Non-CO_2 gases and non-GHG pollutants

Facility-reported CH_4 and N_2O are delivered as reported. Where a facility does not report a gas, its value is derived from the facility's CO_2 using the country-and-sector gas ratio implied by the national inventory. A gas is derived only where the national inventory reports that gas for the country and sector; where the national inventory carries no CH_4 (for example), no facility-level CH_4 is emitted rather than a zero.

Reported non-GHG pollutants (NOx, SOx, NH_3), available from the European registries, are delivered where reported and never modeled for other facilities, since presenting a registry-only population as globally representative would be misleading.

2.1.3.2.7 Reconciliation to the country total

CO_2 is reconciled first. By construction, anchored plus allocated facility CO_2 cannot exceed the national total, because the allocation pool is bounded by the residual. Where reported facility values alone exceed the national total, the exceedance is accepted; a country total can genuinely be too small, and reported values are never modified. Conversely, where the facility sum falls short of the national total, the shortfall is real emissions the sector believes occurred but could not attribute to any known facility. CH_4 and N_2O are then reconciled per gas: reported values are fixed, while derived values may be scaled down where a country's derived gas totals would otherwise exceed its national envelope for that gas.

2.1.3.2.8 Process, combustion, and biogenic CO_2

Each delivered CO_2 value is split into a process percentage and a combustion percentage. Process CO_2 is the non-combustion chemistry term (calcination in lime, carbonate decomposition in glass); for fermentation products (brewing, winemaking), the process share is biogenic fermentation CO_2. Slaughter wastewater CH_4 is excluded from these sectors, as it is accounted for in Climate TRACE's industrial wastewater sector.

2.1.3.3 Asset confidences

Every value in the delivered asset files carries its own confidence rating, in a companion file that is row-for-row and column-for-column parallel to the asset file. Ratings use a five-point vocabulary (very high, high, medium, low, and very low); very high is deliberately never used, because high is reserved for the strongest evidence present in these sectors; nothing here is measured at the stack.

The rating of each field reflects how that specific value was produced: a facility's reported CO_2 for the year is high; a carried-forward or gas-inferred CO2 is medium; an allocated CO_2 is low (with size evidence) or very low (equal share). A derived CH_4 or N_2O is never better than low, because it comes from a national ratio applied to one plant, even on a facility whose CO_2 is reported.

Activity and the emission-factor columns are never better than low, as quotients of an estimated quantity and a library factor.

2.1.3.4 Asset uncertainties

Uncertainties are reported as one absolute 1-sigma uncertainty per value, in the same units as the value itself (tonnes for emissions, tonnes of product for activity). Uncertainties are derived from the same per-field confidence grades. The relative widths are: 10% for a facility's own verified filing for that year; 30% for the facility's own figure carried from another year or inferred from another gas; 40–50% for values built from reported production or capacity with library factors; 75% for allocated values sized by indirect evidence (a size proxy never checked against the plant's actual output); and 100% for equal-share allocations, whose 1-sigma interval reaches zero. The capacity factor, being bounded on [0, 1], carries an absolute rather than relative sigma.

2.2 Implicit Estimation of Other Subsectors

For sectors where there was incomplete or no asset level data, the implicit estimation approach was used, which employed GHG inventories, projected emissions data to fill in missing years in GHG inventories, and developed sector specific equations to output implicit estimation for the sectoral "other" emissions. Sectors estimated using this method are directly equal to their source dataset.

2.2.1 Datasets employed

Climate TRACE GHG emissions inventory data (Climate TRACE 2025; https://climatetrace.org/): Climate TRACE is a global coalition of organizations with each organization focusing on measuring CO_2, N_2O, and CH_4 emissions from particular economic or land cover sectors.

Temporal coverage: Climate TRACE country-level data is provided for 2015-2024 with 2025 emissions either partially or fully projected, depending on temporal coverage for individual sectors. The data are typically released with a 3-4 month latency. For example, data released in November are generally provided through June of the same year, with the remaining months projected.

Sectoral coverage: In addition to the 81% of global emissions covered by Climate TRACE sectors using techniques 1, 2, and 3, 2% is covered by technique 4. The remaining 17% of emissions that are not currently estimated using techniques 1 to 4 were measured using the implicit estimation method. By employing all techniques described here, Climate TRACE country-level inventory accounts for 100% of known anthropogenic greenhouse gasses based on IPCC and UNFCCC reporting.

Geographic coverage: Emissions estimates for Climate TRACE measured sectors are available for every country in the world. Comprehensive data are not available in some disputed territories due to lack of data availability.

Food and Agriculture Organization of the United Nations (FAOSTAT) is a specialized agency of the United Nations that focuses on agriculture and food security. FAO is also tasked with collecting food, agriculture and land use, land-use change and forestry (LULUCF) data from countries around the world regarding their agricultural and forestry activities and emissions.

Temporal coverage: FAOSTAT data is available from 1990 - 2023 at the time of data was accessed (February 23, 2026). Data is released on a two year delay. For example, 2019 data was released in 2021.

Sectoral granularity and coverage: FAOSTAT data is available only for agriculture and LULUCF sectors.

Geographic coverage: FAOSTAT data is global and covers 233 countries and territories (FAOSTAT, 2025).

The Emissions Database for Global Atmospheric Research (EDGAR) is an independent database of anthropogenic GHG emissions, providing both national totals and gridmaps. Though the gridmaps are highly granular, we only used the national totals for the purposes of this exercise.

Temporal coverage: EDGAR data is available for the years 1970-2024 (EDGAR v2025). The complete data is published on an irregular lag. For example, 2018 data was published in May of 2021 and data for 2019, 2020 and 2021 was published in September of 2022. However, EDGAR released data for 2022 in October 2023 and Climate TRACE was able to use these updated values for implicit estimation. EDGAR's 2023 release did not include fluorinated gases broken down by gas. In order to be able to estimate CO_2 equivalent 20- and 100-year global warming potentials (CO_2e 20yr and 100yr GWP), Climate TRACE needed the individual greenhouse gas numbers. For this reason, Climate TRACE utilized EDGAR's previous release (EDGAR v7.0) which provides individual GHGs and forward filled 2021 data to 2024 for fluorinated gases to generate CO_2e 20yr and 100yr GWPs that include all GHGs.

Sectoral granularity: EDGAR data covers all IPCC emissions except for LULUCF. However, the sectors are not as granular as the most granular IPCC categories. For example, EDGAR reports emissions for 1.B.1 Solid Fuels but does not break it down into 1.B.1.a Coal Mining, 1.B.1.b Solid Fuel Transformation, 1.B.1.c Other.

Geographic coverage: EDGAR data is globally comprehensive.

The Community Emissions Data System (CEDS) developed by the Pacific Northwest National Laboratory is an annually updated emissions dataset that estimates anthropogenic emissions for all major emission species (GHGs and non-GHG), providing both country-level and gridded emissions data.

Temporal coverage: CEDS data is available for the years 1750-2023 (v_2025_03_18).

Sectoral granularity: CEDS data covers all IPCC emissions except for LULUCF and biomass burning.

Geographic coverage: CEDS data is globally comprehensive.

2.2.2 Methods

2.2.2.1 Projection approach

Not all GHG inventories have the most recent emissions estimates. EDGAR v2025 currently provides data up through 2024, and FAOSTAT published data up through 2023. In order to utilize the FAOSTAT data for implicit estimation, the data was forward filled up to the final Climate TRACE release month. This method uses the last available estimate as the future projected values. This approach was adopted to ensure all GHG inventories described in section 2.1 match temporally.

2.2.2.2 Implicit Estimation approach - equations

An immediate practical challenge in implicit estimation is that different global emissions inventories do not use one consistent standardized hierarchy of definitions for emitting sectors and subsectors. Thus, Climate TRACE's first step was to define a crosswalk of all emitting subsectors from all inventories it could find, to enable comparison between Climate TRACE sector lead results and those of existing publicly available global emissions inventories. This crosswalk can be found in Table S1 in the Supplementary section.

With the knowledge of what is covered by Climate TRACE vs. EDGAR and FAOSTAT, we were able to write equations to describe the emissions that are not currently estimated using techniques 1, 2, or 3.

Example 1: Other Solid Fuels

Emissions from 'other solid fuels' are not currently measured at the asset level by Climate TRACE. WattTime and Global Energy Monitor are measuring coal mine activities' emissions and TransitionZero is measuring only CO_2 emissions from steel manufacturing, but neither of these models include emissions from solid fuel transformation and abandoned coal mines. Within the IPCC framework, coal mining is a subset of 1.B.1 Solid Fuels. EDGAR reports data for 1.B.1 Solid Fuels but does not report at the more granular levels such as 1.B.1.a Coal Mining and

Handling. Figures 2 and 3 show the WattTime/Global Energy Monitor and EDGAR coverage, respectively for this sector.

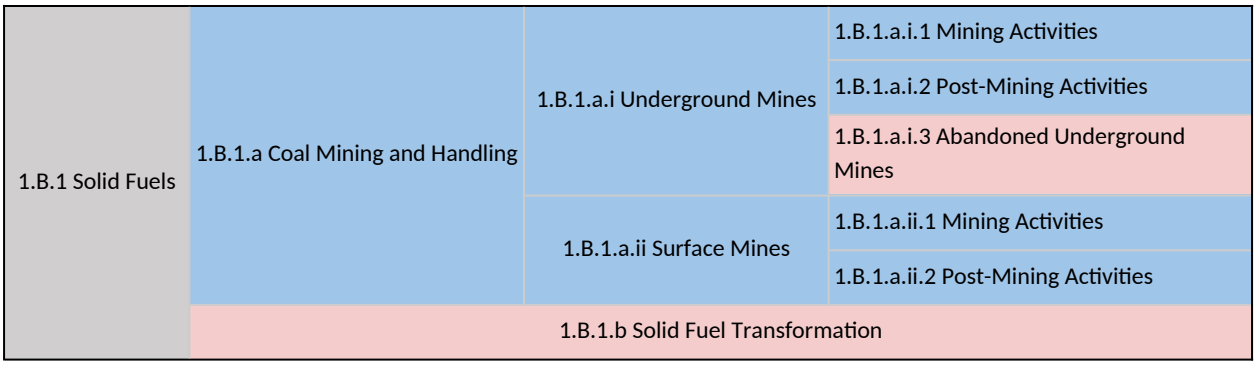


Figure 2 WattTime/Global Energy Monitor 1.B.1 Solid Fuels and subsectors coverage. Fully covered (in blue), not covered (in red), and partially covered (in gray).

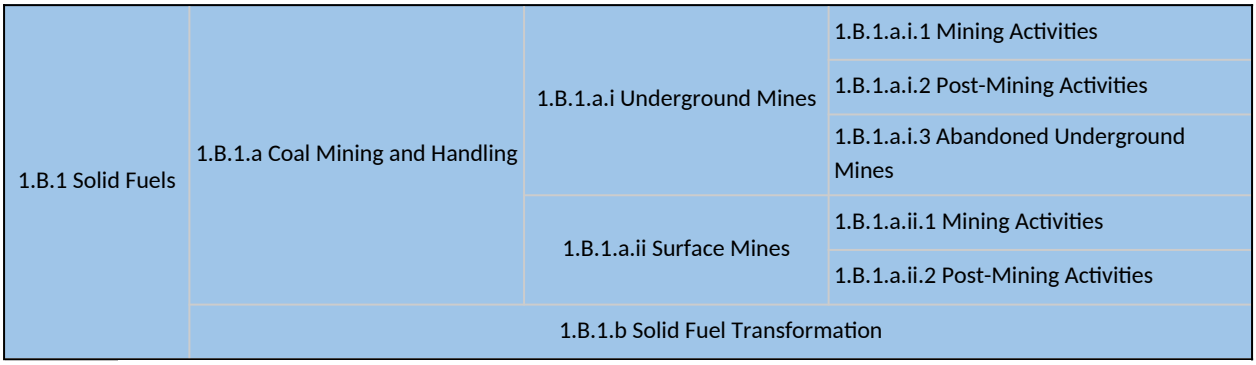


Figure 3 EDGAR 1.B.1 Solid Fuels and subsector coverage. Full covered (in blue), not covered (in red), and partially covered (gray).

Because EDGAR covers all of 1.B.1, subtracting WattTime/Global Energy Monitor's coal mining values from EDGAR results in an estimate for 1.B.1.b Solid Fuel Transformation.

Eq. Climate TRACE Solid Fuel Transformations = EDGAR 1.B.1 Solid Fuels - WattTime/Global Energy Monitor Coal Mining

The approaches highlighted in the examples were repeated for different sectors to estimate remaining emissions. All implicit estimation equations can be found in Table S2 in the Supplementary section.

Example 2: Other agricultural soil emissions

Agricultural soils include N_2O and other emissions from soils due to agricultural practices. While Climate TRACE estimates emissions from some agricultural sectors, like the application of synthetic fertilizers, other emissions from soils were estimated by using FAOSTAT and EDGAR data which reports emissions by subsector and were added together into a single subsector:

$$\begin{aligned} \textit{Other-agricultural-soil-emissions} = {} & \textit{FAOSTAT Manure left on Pasture-non cattle} + \\ & \textit{FAOSTAT Drained organic soils} + \\ & \textit{EDGAR 3.C.2 Liming} + \\ & \textit{EDGAR 3.C.3 Urea application} \end{aligned}$$

Example 3: Other Manufacturing

Compared to the v3.0.0 Climate TRACE release in 2023, the Other Manufacturing sector in v5.0.0 contains fewer subsectors, as they were further granularized either through technique 4 or through the introduction of the CEDS inventory (see Section 3.1 below and Table S2). Additionally, the remainder of the combustion (i.e. indirect) emissions from non-metallic minerals (1.A.2.f) described in Sections 2.1.3.1.2 and 2.1.3.1.3 must be accounted for here. This was done using the same formulation above and is denoted in the following equations as '*misc-combustion*'. Therefore, the equation for estimating the remaining, or 'other', manufacturing emissions is:

$$\begin{aligned} \textit{other-manufacturing} = {} & \textit{EDGAR 2.D non-energy products} + \\ & \textit{(EDGAR 2.A.X Other minerals - CEDS 2.A.3 Glass)} + \\ & \textit{CEDS 1A2g_Ind-Comb-Construction} + \\ & \textit{CEDS 1A2g_Ind-Comb-transpequip} + \\ & \textit{CEDS 1A2g_Ind-Comb-machinery} + \\ & \textit{CEDS 1A2g_Ind-Comb-other} + \textit{misc-combustion} \end{aligned}$$

where,

$$\begin{aligned} \textit{misc-combustion} = {} & \textit{(CEDS 1A2f_Ind-Comb-Non-metalic-minerals)} \times \\ & \textit{EDGAR 2.A.4 Other Process Uses of Carbonates} / \\ & \textit{(EDGAR 2.A.1 Cement} + \textit{EDGAR 2.A.2 Lime} + \\ & \textit{EDGAR 2.A.3 Glass} + \\ & \textit{EDGAR 2.A.4 Other Process Uses of Carbonates)} \end{aligned}$$

The Other Manufacturing sector derived here contains the following subsectors: combustion related to construction, transportation equipment, machinery, non-metallic mineral production

(excluding glass, lime, and cement), as well as all other combustion (1.A.2.g), and non-energy products (2.D).

3. Results

The results below highlight the disaggregation and implicit estimation approaches outlined above. Unless otherwise noted, all CO_2 equivalent (CO_2e) estimates are in the 100 year timescale (AR6 global warming potentials). These results are based on the V5.11.0 dataset delivered September 2026.

3.1 Data-informed Emissions Disaggregation

Table 1 summarizes the total global GHG emissions from the disaggregated manufacturing sectors, which were all previously (v3.0.0) combined into the *other-manufacturing* sector. Overall, more than 50% of emissions, or roughly 2 billion tonnes CO_2e in 2024, previously categorized as *other-manufacturing* were disaggregated into these subsectors.

For the four rebuilt sectors, the share of each sector's national totals attributed to named facilities in 2025 was 83.2% for *glass*, 81.8% for *textiles-leather-apparel*, 61.5% for *lime*, and 48.8% for *food-beverage-tobacco*; the remaining emissions could not be attributed to a specific site, and are therefore reported as spatially uncertain emissions rather than being spread across known facilities. The delivered asset files cover roughly 54,000 facilities across these four sectors.

Finally, the top five highest emitting countries in each sector are presented in Table 3. In every sector other than *other-metals* either China or the USA is the highest emitter, and in five of the nine sectors, these two countries hold the top two spots. In fact, there are only two sectors where either China or the USA falls outside the top five emitting countries (textiles-leather-apparel, USA; glass, China). The remaining high-emitting spots are largely sector-dependent, with Indonesia, for instance landing in the top-five for *other-metals* sectors, and *glass* being largely dominated by European countries, following the USA.

Table 1. Global totals of CO_2e emissions across the nine manufacturing subsectors, including the percent contribution of each subsector to the total global for all. Note, total % contribution may not equal 100 due to rounding.

Sector	2025 Emissions [Mt CO_2e]	% contribution
food-beverage-tobacco	259	7.0
glass	31	0.8
lime	528	14.2
other-chemicals	796	21.4

Sector	2025 Emissions [Mt CO_2e]	% contribution
other-manufacturing	1,915	51.5
other-metals	82	2.2
other-mining-quarrying	43	1.1
textiles-leather-apparel	50	1.3
wood-and-wood-products	15	0.4
Total	3,718	100

Table 2. Share of 2025 GHG emissions per sector attributed to named facilities.

Sector	2025 asset-level emissions [Mt CO_2e]	% of sector total attributed to named facilities
food-beverage-tobacco	126	49
glass	26	83
lime	324	61
other-chemicals	89	11
other-manufacturing	17	1
other-metals	38	46
textiles-leather-apparel	41	82

Table 3. Highest five GHG-emitting countries for each manufacturing subsector in 2025.

Country	*2025 Emissions [Mt CO_2e]*
other-mining-quarrying	
China	19.6
USA	3.4
South Africa	2.8

Country	*2025 Emissions [Mt CO_2e]*
other-chemicals	
China	176.1
USA	165.7
Russia	45.3

Country	*2025 Emissions [Mt CO_2e]*
India	1.8
Canada	1.6
other-metals	
Indonesia	23.0
USA	20.1
South Africa	8.0
Norway	5.4
France	5.0
food-beverage-tobacco	
USA	88.7
China	50.6
Germany	8.3
Japan	7.8
France	6.5
lime	
China	385.0
USA	27.4
India	18.2
Russia	14.0
Brazil	9.1

Country	*2025 Emissions [Mt CO_2e]*
Japan	43.0
Iran	35.8
wood-and-wood-products	
USA	4.9
China	2.3
Canada	1.8
Russia	1.4
Germany	0.5
textiles-leather-apparel	
China	18.9
India	4.5
Turkey	4.3
Argentina	2.6
South Korea	2.0
glass	
USA	7.5
Great Britain	3.5
Russia	3.1
Italy	2.5
Germany	2.5

Country	*2025 Emissions [Mt CO_2e]*
other-manufacturing	

China	632.2
India	186.7
USA	129.4
Iran	114.7
Saudi Arabia	106.1

3.2 Implicit Estimation of All Other Subsectors

The goal of the implicit estimation technique is to complement Climate TRACE's estimates using techniques 1, 2, 3 and 4 with existing estimates. This serves two purposes:

1) Ensure Climate TRACE's end product is globally and sectorally comprehensive for ease of use.
2) Make Climate TRACE and other datasets interoperable enough that they can be compared statistically.

Figure 4 shows EDGAR and Climate TRACE global totals for 2015-2024. Climate TRACE totals are split into implicitly estimated and measured totals. These totals include only positive differences from the implicit method (see Section 3.2.2 and Table 6). It should be noted that the Climate TRACE's global estimates are between 3.8 and 6.8 Gigatonnes higher than EDGAR's for all years. This is due primarily to 1) RMI's oil and gas estimates which indicate vast undercounting of methane in current oil and gas estimates and 2) inclusion of non-fossil fuel emissions in the Climate TRACE emissions totals. Table 4 provides the EDGAR and Climate TRACE emission differences.

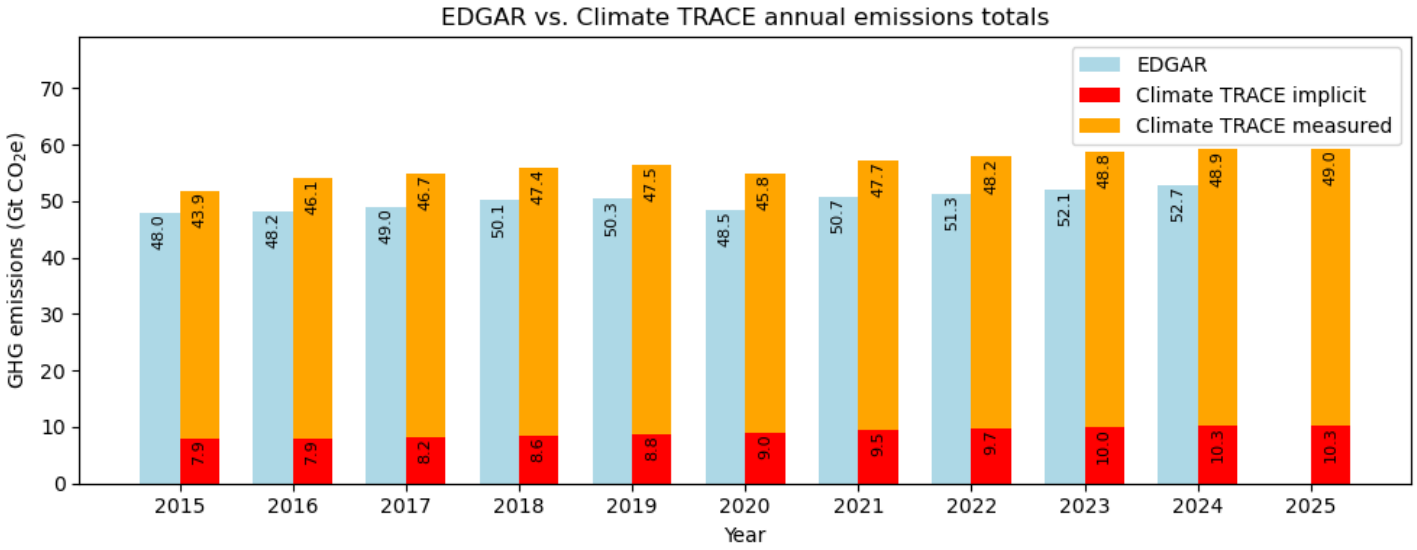


Figure 4 EDGAR global totals compared to Climate TRACE global totals for years 2015 to 2025. Climate TRACE totals are split into implicitly estimated and all other categories. Blue bars = EDGAR Totals, Red bars = Climate TRACE implicit estimation estimates, and Orange bars = Climate TRACE measured.

Table 4 Climate TRACE V5.11.0 global totals versus EDGAR 2025 global totals, and the difference between the two by year. Emissions provided in Gigatonnes (GT).

Year	Global Climate TRACE emissions (GT CO_2e)	EDGAR CO_2e Emissions (GT)	Difference (GT CO_2e)
2015	51.8	48.0	3.8
2016	54.1	48.2	5.9
2017	54.9	49.0	5.9
2018	56.0	50.1	5.9
2019	56.3	50.3	6.0
2020	54.8	48.5	6.3
2021	57.2	50.7	6.5
2022	57.8	51.3	6.6
2023	58.8	52.1	6.8

2024	59.2	52.7	6.4
2025	59.3	N/A	N/A

Figure 5 shows EDGAR 2025 totals compared to Climate TRACE V5.11.0 totals for the top 10 highest emitting countries for 2024. China is the highest emitter, ~17.4 Gigatonnes, which is ~2.5 times more emissions than the USA (Table 5). Generally, Climate TRACE total estimates (measured and implicit estimate) result in higher emissions relative to EDGAR emissions.

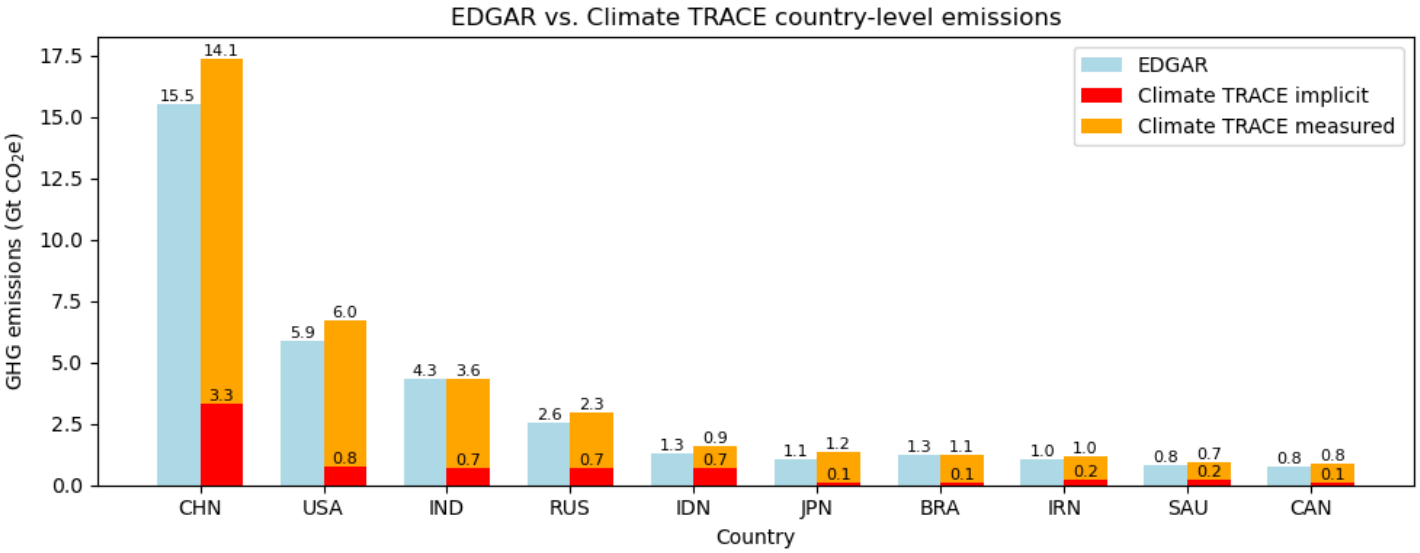


Figure 5 EDGAR 2025 totals compared to Climate TRACE V5.11.0 totals for the top 10 highest emitting countries for 2024. Blue bars = EDGAR Totals, Red bars = Climate TRACE implicit, and Orange bars = Climate TRACE measured.

Table 5 EDGAR 2025 totals compared to Climate TRACE V5.11.0 totals for the top 10 most emitting countries in 2024. Emissions provided in Gigatonnes (GT).

Country	EDGAR Total (GT CO_2e)	Climate TRACE Total (GT CO_2e)	Climate TRACE percent using techniques 1, 2,3 or 4	Climate TRACE percent implicitly estimated
CHN	15.5	17.4	81	19
USA	5.9	6.7	89	11
IND	4.3	4.3	83	17
RUS	2.6	3.0	77	23
IDN	1.3	1.6	58	42

Country	EDGAR Total (GT CO_2e)	Climate TRACE Total (GT CO_2e)	Climate TRACE percent using techniques 1, 2,3 or 4	Climate TRACE percent implicitly estimated
JPN	1.1	1.3	91	9
BRA	1.3	1.3	91	9
IRN	1.0	1.2	83	17
SAU	0.8	0.9	77	23
CAN	0.8	0.9	87	13

3.2.1 Verifying results

We went through several systematic audits of the results to ensure that the implicit estimation equations were properly applied using all available relevant data. For instance, due to differences in how inventories specify country names, we ensured that standardized International Organization for Standardization (ISO3) codes were used to match up data points for each implicit calculation.

Results were not reported in cases where inventories define the actual emitting country differently. For example, EDGAR reported data for Serbia and Montenegro as one country. Climate TRACE and FAOSTAT report them separately since they have become separate legal entities since 2006. Therefore, results relying on EDGAR data for such countries were excluded from reported results. Similarly, results for some sectors are not included for countries in the Climate TRACE dataset where EDGAR and/or FAOSTAT do not provide emissions estimates.

3.2.2 Negative results from the implicit estimation approach

In some cases, implicit estimation equations resulted in negative emissions estimates. We identified two main reasons for this:

1. There is a large disparity between emissions estimates from different inventories for the relevant sectors. This is particularly the case for sectors like ‘other-chemicals’, ‘other-metals’, and ‘other-solid-fuels’, because Climate TRACE’s emissions for subsectors like methanol, ammonia, and aluminum manufacturing are significantly higher than EDGAR/CEDS’ emissions estimates for the overall sector resulting in negative emissions. EDGAR and CEDS rely, in part, on fuel consumption data from the International Energy Agency while Climate TRACE’s approach is facility level focused.
2. Data from relevant sectors are missing from one of the inventories.

Examining the causes of negative results can identify discrepancies that are relevant to comparing estimates between inventories. Overall, 6.9% of implicit estimation results were negative in the dataset. However, they were distributed across just 6 out of the 19 sectors in the analysis. These results are highlighted in Tables 6 and 7. Climate TRACE replaces the negative numbers with 0 in the final inventory.

Table 6 Sum of negative implicit estimate results by sector (tonnes CO_2e) in V5.11.0 release.

Year	Other Chemicals	Other Energy Use	Other Manufacturing	Other Metals	Other Mining Quarrying	Other Solid Fuels
2015	-3,456,104	-458,021,258	-477	-2,237,258,181	-24,275,115	-629,760,341
2016	-3,288,209	-415,374,380	-475	-2,259,698,646	-25,114,046	-740,193,156
2017	-3,552,864	-399,011,936	-474	-2,363,900,240	-23,146,589	-731,624,942
2018	-2,985,914	-436,512,163	-472	-2,444,458,029	-22,832,225	-681,618,833
2019	-3,803,155	-394,515,457	-470	-2,536,530,085	-23,348,134	-661,250,146
2020	-2,190,113	-394,774,037	-517	-2,603,414,525	-71,068,101	-678,135,822
2021	-3,569,225	-346,017,379	-579	-2,660,625,561	-64,715,586	-787,429,329
2022	-3,417,698	-358,009,497	-579	-2,586,079,978	-77,014,180	-799,406,661
2023	-3,930,914	-341,557,454	-579	-2,576,388,855	-79,413,011	-778,796,429

Table 7 Number of countries with negative implicit estimate results by sector in V5.11.0 release.

Year	Other Chemicals	Other Energy Use	Other Manufacturing	Other Metals	Other Mining Quarrying	Other Solid Fuels
2015	14	142	3	86	78	39
2016	14	146	3	86	80	39
2017	15	145	3	83	80	38
2018	14	141	3	83	77	37
2019	14	143	3	84	75	37
2020	11	146	3	85	80	36
2021	12	147	3	85	82	34
2022	13	144	3	84	83	37
2023	13	142	3	85	83	37

4. Implications and future work

The two methods outlined here have many strengths, namely:

1. **Inventory completeness**: Although Climate TRACE's long-term goal is to estimate all emissions down to the source/asset, these approaches ensure that no emissions are being excluded from country/ sector totals;
2. **Estimates for low-information facilities**: Many manufacturing sector facilities– particularly those in countries without extensive publicly available databases– have little to no emissions data available at the facility level. The data informed disaggregation approach utilizes available information to create a first-order emissions estimate. As more data are gathered – such as more granular economic data or additional information about the facilities, these estimates can be iteratively improved.
3. **Novel inventory crosswalk**. By defining all third party GHG inventory sectors by their IPCC reporting categories, Climate TRACE was able to make comparisons that would have otherwise been difficult. These comparisons also identify countries/ sectors where there are the largest discrepancies among all the inventories;
4. **Combining multiple estimates increases trust**. Every technique for estimating GHGs, especially at a global level has pros and cons. Some are highly granular and accurate but can be expensive (such as ground-level monitoring from CEMS units) and not comprehensive. This implicit estimation approach created by Climate TRACE for the 2023 launch lays the groundwork for more extensive Bayesian metamodeling that combines many sources of information to improve emissions estimates.

4.1 Limitations of Data-informed Emissions Disaggregation

The purpose of this work was to disaggregate, at multiple levels, existing emissions inventories: first from combined inventory categories into individual manufacturing subsectors (Section 2.1.3.1), and second from national totals down to individual facilities (Section 2.1.3.2). There are limitations to the framework presented:

1. Most emissions are allocated, not measured. Outside the countries with mandatory facility-level reporting, the large majority of facility values are one country's sectoral total divided across the facilities the project could locate, in proportion to indirect evidence of relative size. The per-value confidence and uncertainty companions exist precisely so that a consumer can distinguish a facility's own regulator-reported figure from an allocated share; allocated values carry relative uncertainties of 75–100%.
2. Production activity is rarely reported. No facility in the assembled population reports its production in mass units, so nearly all activity values are derived from emissions and an emission factor rather than observed. Lime is the exception, where national production statistics provide an independent anchor. Obtaining more accurate and granular production data at the country level is a primary goal of future iterations.
3. Facility coverage is incomplete and uneven. Although this dataset contains the greatest number of emitting assets for the subsectors presented here, asset-level information in most countries is still lacking, and the facilities visible in registries are systematically the larger ones. The completeness denominator (national establishment counts) makes this gap

measurable per country, and the unattributed remainder of each national total is reported rather than forced onto known facilities.

4. Upstream publisher data is accepted with flags, not corrected. Reported values are never modified, including a small number of implausible filings (for example, facility-years whose reported emissions exceed their own national sector total, or single-year spikes in a facility's series).
5. Small-geography artifacts. Where an overseas territory or very small country hosts a single real facility against an island-sized national envelope, one facility can legitimately constitute the entire sector, and ratio-based diagnostics there reflect geography rather than error.

Addressing each of these limitations in future iterations of this dataset will be of primary interest to increasing its completeness and improving its accuracy.

4.2 Limitations of the implicit estimation approach

1. **Low confidence estimates.** This method makes use of estimates that are not based on direct observations. Climate TRACE's roadmap will be to downweigh such estimates in future metamodeling and upweight estimates based on direct observations.
2. **Novel estimates.** In sectors containing one or more implicitly estimated subsectors, any potential errors in the source data potentially affects the sector total, not just the implicitly estimated subsector total. As long as the sum of subsectors measured using techniques 1, 2, and 3 (from the introduction) are less than the sum of the implicitly measured subsectors, then the sector total will match exactly the source data used to implicitly estimate the remaining sectoral "other" emissions (Table 7). For example, the sum of Climate TRACE's chemicals and manufacturing sector estimates is almost always lower than the sum of the corresponding EDGAR subsectors. The implicitly estimated sector, Other Manufacturing, will always cause the sum of all Climate TRACE manufacturing sectors to equal the sum of EDGAR's manufacturing sectors. Therefore, this method is useful for detecting underestimates in the original source data. However, if the estimated total is an overestimate, it may not correctly detect it.
3. **Lack of specificity in implicitly estimated sectors.** This method does not reveal emitting source level results or identify the exact cause of emissions in implicitly estimated sectors.
4. **Reliability of source datasets.** EDGAR generally publishes estimates on a lag of 1 year. EDGAR has been annually publishing updates in the fall of every year since. However, the EDGAR release schedule is not published to the public. In the absence of up-to-date datasets for implicit estimation, Climate TRACE will need to create projections to fill in missing years using the most recently available data.
   Likewise, the most recent published data from FAOSTAT is currently 2023 at the time of this work. Climate TRACE forward fills this data for the 2023 to 2025 estimates.

Thus, Climate TRACE's priorities for future implicit estimation are to incorporate as many emissions inventories as possible, produce similar systems of equations to make them interoperable as possible, and produce parsimonious consensus results based on large numbers of independent cross-validations.

5. Emission Reduction Solutions

Identifying ways to reduce emissions is a key goal for Climate TRACE. Here, Table 8 provides a summary of potential emission reduction solutions (ERSs), and descriptions are provided for each sector estimated in this methodology to understand how mitigation strategies can reduce emissions.

Table 8 Emission reduction solutions (ERS) by sector. Values are the fraction of each GHG's emissions removed by the strategy (0.30 = 30% reduction). A hyphen (-) means the strategy does not reduce that gas. NA means no value is given in the text. Section numbers in the notes refer to subsections of Section 5.

Sector	Strategy name	CO2	CH4	N2O	F-gases	Notes
biological-treatment-of-solid-waste-and-biogenic	CH4 flaring	-	0.36	-	-	40% digested x 90% flare efficiency; gas not named, treated as CH4 (5.1)
food-beverage-tobacco	Thermal management and waste heat recovery	0.30	-	-	-	CO2 EF ratio 0.7, applied to all activity; induced electricity 0.44 (5.2.1)
	Switch to biomass boilers	0.10	-	-	-	Applied to fossil CO2 EF; low confidence (5.2.2)
glass	Convert production configuration (full electrification)	0.55	-	-	-	CO2 EF ratio 0.45; container glass and unknown type only; induced electricity 0.9 (5.3.1)
	Increase glass recycling	0.30	0.15	0.15	-	EF ratios 0.7 (CO2) and 0.85 (CH4, N2O); 50% cullet (5.3.2)
incineration-and-open-burning-of-waste	Divert organic waste from landfills and compost/capture	0.85	0.85	0.85	-	Text gives about 85% without naming gases; applied to all three (5.4.1)
	Domestic waste management infrastructure	1.00	1.00	1.00	-	Activity 'effectively eliminated'; no number given, 1.00 inferred (5.4.2)
lime	Energy from biomass	0.30	-	-	-	About 30% overall; gases not named, treated as CO2 (5.5)
manure-management-other	Slurry removal system	-	0.65	0.12	-	Lower end of 65-80% (CH4) and 12-30% (N2O) (5.6)
other-agricultural-soil-emissions	Enhanced rock weathering (ERW)	0.20	-	-	-	Net of electricity; 0.35 gross; N2O not considered (5.7)
other-chemicals	Alternative fuel use and cleaner production	0.30	0.30	0.30	-	About 30% overall (15% + more than 15%); gases not named, applied to all three (5.8)
railways	Electrification of railways	0.40	0.40	0.40	-	About 40% net of electricity generation; gases not named, applied to all three (5.9)
fluorinated-gases	Replace HFCs and HCFCs with low GWP alternatives	-	-	-	0.70	Average of 50-90%; covers about 90% of HFC usage (5.10.1)

	Lifecycle refrigerant management	-	-	-	0.20	EF ratio 0.8 on 60% of activity, about 0.12 overall; low confidence (5.10.2)
other-mining-quarrying	Electrification of mine equipment	NA	NA	NA	-	No percentage given; 'fully eliminate' at some sites only (5.11)
textiles-leather-apparel	Energy efficiency + heat pumps: wet processing	0.30	-	-	-	EF ratio 0.70; induced electricity 3.29 (5.12)
	Energy efficiency + heat pumps: textile production	0.10	-	-	-	EF ratio 0.90; induced electricity 1.49 (5.12)
	Energy efficiency + heat pumps: garment and product manufacturing	0.10	-	-	-	EF ratio 0.90; induced electricity 0.60 (5.12)
	Energy efficiency + heat pumps: general (type unknown)	0.15	-	-	-	EF ratio 0.85; induced electricity 2.40 (5.12)
wood-and-wood-products	Energy from biomass	0.66	-	-	-	'Up to' 66% net emissions; gases not named, treated as CO2 (5.13)
enteric-fermentation-other	Feed additives in ruminant diet	-	0.20	-	-	Around 20% of enteric CH4 (5.14)
cropland-fires	Residue removal without burning	1.00	1.00	1.00	-	Burning emissions eliminated; no number given, 1.00 inferred (5.15)

5.1 biological-treatment-of-solid-waste-and-biogenic - CH_4 Flaring

This proposed ERS involves collecting methane produced through anaerobic digestion and flaring it, thereby converting CH_4 into CO_2. Because methane has a higher global warming potential than CO_2, flaring reduces the overall climate impact of waste management. Of the waste diverted for biological treatment, around 40% could be anaerobically digested and captured for flaring. Flaring systems operate at approximately 90% efficiency, which translates to a potential total emissions reduction of roughly 36% from the treated waste stream. This methodology accounts for both the methane emissions reduction and the associated mitigation of greenhouse gas warming potential.

Many countries and landfills already employ methane flaring to limit the warming impact of organic waste decomposition. While the approach is effective, technological and financial constraints may hinder global adoption, particularly in regions lacking infrastructure or funding for anaerobic digestion and flaring systems.

5.2 food-beverage-tobacco

5.2.1 Thermal management and waste heat recovery

Process heat accounts for about 60–70% of total energy needs at many food processing facilities, and many of these heating processes are inefficient (Sovacool et al., 2021). Thermal management and waste heat recovery target this energy use through improved insulation, better recovery of waste heat (for example in baking), and “pinch technology”, which optimizes heat exchange where

hot and cold streams are close enough in temperature that their composite curves meet. Reported savings vary widely by process: pinch analysis has cut energy requirements by 24% in brewing and by up to 51% in milk powder production, and improved insulation and design can reduce the energy used by heat exchangers in biscuit manufacturing by 45%. The review also reports CO_2 emissions reductions of 5–35% in Swedish meat plants using heat exchanger networks and heat pumps, and energy savings of 35–95% for selected dairy operations (Sovacool et al., 2021).

These measures reduce heat losses and reuse heat that would otherwise be wasted, lowering the fuel needed for processing. Given the variation in reported outcomes and process boundaries, an average reduction of 30% is assumed for modeling purposes. This is represented by a CO_2 emissions factor ratio of 0.7 applied to all activity, with low confidence. Where recovered heat is upgraded with electrically driven equipment, the strategy induces additional electricity demand, which is accounted for in the electricity-generation sector (induced activity conversion rate of 0.44).

Many food, beverage and tobacco producers have already adopted thermal management and waste heat recovery, driven mainly by energy costs; industry sources report energy savings of up to about 25% with payback periods of three to five years (Fenton, 2024). Global adoption may be limited by how much the reduction potential varies from process to process, but the energy cost savings these measures offer can help drive wider uptake.

5.2.2 Switch to biomass boilers

Boilers that burn coal, oil or natural gas supply much of the steam and hot water used in food, beverage and tobacco processing. Replacing them with biomass-only boilers, or with combined biomass and fossil-fuel boilers, reduces fossil fuel use for heating. Potentially sustainable biomass includes agricultural residues and biogas from the anaerobic digestion of wastewater and organic process waste. At a large distillery, for example, anaerobic digestion of by-products could meet up to 64% of natural gas demand and cut direct greenhouse gas emissions by 54% (O'Shea et al., 2020).

This strategy assumes an average 10% reduction in emissions, applied to the fossil CO_2 emissions factor, with low confidence. Reductions can be considerably higher where sustainable biomass replaces coal, or where most fossil natural gas is displaced. Biomass boilers are being widely adopted in food processing in New Zealand, where they are replacing coal-fired boilers. Wider uptake depends on a reliable local supply of biomass or digester gas and on the cost of boiler conversion.

5.3 glass

5.3.1 Convert production configuration (full electrification)

The main source of emissions in glass manufacturing is the combustion of fossil fuels to reach the very high temperatures needed to melt the batch. Converting furnaces to full electric melting, in which the glass is heated directly by electrodes, removes most on-site fuel combustion. Electric melting is one of several industrial technologies available to reduce CO_2 from glass furnaces (Atzori et al., 2023). A recent techno-economic assessment found that an all-electric furnace reduces emissions by 41% compared with a conventional gas-fired furnace, once the emissions from the electricity it uses are included (Salman et al., 2025).

The strategy is modeled as a 55% reduction in direct emissions, represented by a CO_2 emissions factor ratio of 0.45, applied to container-glass assets and to assets of unknown type, with low confidence. Emissions from the additional electricity demand are accounted for in the electricity-generation sector (induced activity conversion rate of 0.9), so the net reduction is smaller than the direct reduction.

Electric melting is most effective at small and medium facilities. All-electric furnaces currently operate mainly at about 10–100 tonnes per day, with some reaching 180 tonnes per day, while large container and flat glass furnaces cannot yet be heated purely by electricity; larger electric furnaces are still being trialed (FfE, 2025). Electrification has been adopted at a number of facilities in Europe. The main barriers are cost, furnace size and the capacity of the local electricity supply.

5.3.2 Increase glass recycling

Glass manufacturing emits CO_2 from two sources in the melting stage: the fuel burned to heat the furnace, and the thermal decomposition (calcination) of carbonate raw materials such as soda ash, limestone and dolomite, which chemically releases CO_2 as the batch melts. Replacing 50% of the virgin feedstock with cullet (crushed recycled glass) reduces both. Because cullet has already been melted, remelting it releases no new carbonate process CO_2. Because it melts more readily than raw batch, the furnace needs less energy: each additional 10% of cullet reduces furnace energy consumption by about 2.5–3% (CTCN, 2016).

By adding 50% cullet, the strategy can reduce the process chemical emissions by 50% and reduce fuel use as well, reducing emission by 15%. This is represented by emissions factor ratios of 0.7 for CO_2 and 0.85 for CH_4 and N_2O, applied to all activity, with low confidence. The reduction assumes the facility currently uses 100% virgin feedstock; plants already using cullet will see a smaller reduction.

European glass furnaces routinely run on 50–60% cullet, and some exceed 90%, so the target is feasible and in places already exceeded. The main barriers are cullet availability and quality: collection volumes are one constraint and purity another, with flat glass requiring the highest purity. Glass packaging collection for recycling in the European Union reached a record 82.2% in

2024, supported by separate municipal collection systems (Nutbean, 2026). The strategy therefore depends mainly on collecting and sorting recycled glass, since most furnaces can adapt to different feedstock.

5.4 incineration-and-open-burning-of-waste

The incineration-and-open-burning-of-waste produces GHGs and non-GHGs, contributing to global emissions and impacting local health. Here, two ERS strategies are proposed for this sector.

5.4.1 Divert organic waste from landfills and compost/capture

Open burning of waste is common in regions with limited waste management infrastructure, and waste incineration is both inefficient and a source of pollutants harmful to human health. A solution to reduce these emissions is to sort waste, diverting organic materials to composting or capturing methane, and separating recyclables from landfill streams. By diverting organic waste to composting, waste incineration emissions can be reduced by approximately 85%. Many communities have already implemented composting programs and methane capture initiatives, demonstrating the feasibility of this approach. However, technological and financial constraints may limit global adoption, particularly in regions lacking the necessary infrastructure or resources (Environment America, 2025).

5.4.2 Domestic waste management infrastructure

Open burning of waste is prevalent in regions with inadequate waste management infrastructure, and waste incineration is inefficient while releasing pollutants that harm human health. A practical solution is the adoption of decentralized domestic waste management practices, including neighborhood-level composting and recycling projects, which can significantly reduce emissions from open burning. By implementing collection and processing programs for recycling and composting, waste incineration activity can be effectively eliminated. Many communities already operate such programs, demonstrating feasibility, but financial constraints may limit global adoption (Environment America, 2025).

5.5 lime - Energy from biomass

Lime production generates greenhouse gas and other gaseous emissions primarily during the calcination of limestone, where heating in kilns releases CO_2 both from fuel combustion and the chemical decomposition of calcium carbonate, contributing roughly 0.9% of global CO2-equivalent emissions. One effective strategy to reduce these emissions is the use of biomass for energy needs during production, replacing fossil fuel combustion, which is generally more emissions-intensive. By substituting fossil fuels with biomass, overall emissions from lime production can be reduced by approximately 30%. This approach has a high technology readiness

level (TRL 8), indicating that fuel conversion to biomass is a mature and implementable method within the industry (Shen et al., 2024).

5.6 manure-management-other - Slurry removal system

Methane and nitrous oxide emissions from manure management practices not included in other sub-sectors account for nearly 0.4% of global CO_2-equivalent emissions. A strategy to reduce these emissions is the implementation of improved slurry removal systems, where slurry is removed from animal housing more frequently. By employing this strategy, methane emissions can decrease by 65–80% and nitrous oxide emissions by 12–30%, with calculations for this methodology assuming reductions at the lower end of these ranges. Technologically, this approach is feasible as it does not require complex new systems, though adoption has so far been tested only in limited settings. The strategy can be integrated into existing manure management operations to reduce greenhouse gas emissions without significant infrastructural changes (Aneja et al., 2023; von Keyserlingk et al., 2023).

5.7 other-agricultural-soil-emissions - Enhanced rock weathering (ERW)

Agricultural soils, excluding fertilization and manure application, release CO_2, CH_4, and N_2O through microbial processes including the decomposition of organic matter and wetting-drying cycles. These emissions contribute approximately 2.2% of global CO_2-equivalent emissions. Enhanced rock weathering (ERW) is a strategy that applies finely ground silicate minerals to soils, where they react with CO_2 in the soil and atmosphere to form stable carbonates. This process lowers soil CO_2 levels and can indirectly improve soil pH and nutrient balance, potentially mitigating conditions favorable to greenhouse gas emissions, though reductions in N_2O are not considered here. Using the emission source curve provided in the literature, ERW is estimated to reduce annual emissions by roughly 35%; when accounting for emissions related to electricity used to produce the minerals, the net reduction is approximately 20% relative to the baseline. While ERW shows potential for substantial impact, it remains a promising but underexplored approach (Beerling et al., 2024).

5.8 other-chemicals - Alternative fuel use and cleaner production

Chemical production of hydrogen, nitric acid, titanium oxide, and other minor chemicals generates greenhouse gas emissions and other pollutants primarily through fuel combustion and process reactions involving oxidation or reforming steps. These processes release CO_2, N_2O, and small amounts of other pollutants directly from reactors, reformers, or kilns used in chemical conversion and purification. Emissions can be mitigated through the use of alternative energy sources such as waste heat recovery and renewable energy, alongside efficiency improvements, as well as cleaner production processes including methane leak reduction and carbon capture. Implementation of alternative energy sources reduces emissions by approximately 15%, while cleaner production

processes further reduce emissions by over 15%, resulting in an overall estimated reduction of around 30%. These practices have been widely adopted and have demonstrated improvements in operational efficiency and emissions reductions (RMI, 2025).

5.9 railways - Electrification of railways

Railways are a source of GHG emissions through fossil-fuel usage, accounting for approximately 0.2% of global CO_2-equivalent emissions. Emissions can be reduced by increasing railway electrification, leveraging historical trends of electrification and greater usage of electrified railways for freight. Implementing this strategy could decrease greenhouse gas emissions by roughly 40%, including the emissions associated with electricity generation. Electrification of railways has been widely adopted, particularly for passenger transport, and many countries plan to expand the use of electrified rail for freight as well (Drawdown, 2025).

5.10 fluorinated-gases

5.10.1 Replace HFCs and HCFCs with low GWP alternatives

Replacing hydrofluorocarbons (HFCs) and hydrochlorofluorocarbons (HCFCs) with low-global warming potential (GWP) alternatives targets fluorinated gases, which, although present in relatively small atmospheric concentrations, have extremely high GWPs and long atmospheric lifetimes, making them some of the most potent anthropogenic greenhouse gases. HFCs are predominantly used as refrigerants in residential, commercial, and industrial air conditioning and refrigeration systems, and are also found in foam production, aerosol sprays, fire suppression systems, and as solvents. The proposed solution involves substituting these high-GWP compounds with natural refrigerants or low-GWP synthetic alternatives, which significantly lowers emissions from these systems.

This strategy addresses approximately 90% of HFC usage in residential, commercial, and industrial applications, focusing particularly on the 86% derived from refrigeration, air conditioning, and heat pump systems. Adoption of low-GWP alternatives, such as hydrocarbons, carbon dioxide, hydrofluoroolefins (HFOs), or blended refrigerants like R32 and R454B, can reduce GWP-weighted emissions by 50–90%, with a conservative average reduction of 70% assumed for this analysis. The mechanism relies on either retrofitting existing equipment or replacing systems at the end of their operational life, ensuring that emissions from these high-impact sources are minimized.

Adoption of low-GWP refrigerants is expanding globally, driven by regulations such as the Kigali Amendment and sector-specific emissions reduction targets. Implementation has been effective in commercial and industrial refrigeration, with innovations in recent years accelerating the deployment of these alternatives. Nonetheless, practical adoption requires careful assessment for

each application due to variations in thermodynamic and safety properties. Local climate, regional conditions, and system-specific constraints can influence the choice of refrigerant, meaning there is no universal solution applicable to all contexts (European Commission, 2025; UNEP, 2015; EPA, 2025).

5.10.2 Lifecycle refrigerant management

Fluorinated gases used as refrigerants enter the atmosphere when they leak from refrigeration and air-conditioning systems through poor installation and maintenance or wear and tear, and when they are not recovered at an appliance's end of life. Residential systems can leak 5–20% of their charge each year depending on system type and location, and most of an appliance's lifetime emissions can occur at disposal if the refrigerant is simply vented. Lifecycle refrigerant management (LRM) addresses each stage (UNEP TEAP, 2024):

- Smart design that minimizes refrigerant charge, such as short piping runs and compact condenser and receiver layouts;
- Routine maintenance and leak detection, from low-cost annual leak tests to automated leak detection in commercial systems;
- Recovery, reclamation and reuse of refrigerant; and
- Destruction of refrigerant that cannot be reused, at end of life.

A 20% reduction in leakage and end-of-life emissions is assumed, applied to 60% of the sector's activity. This is represented by an emissions factor ratio of 0.8 on 60% of activity, about a 12% reduction in sector emissions overall. The effectiveness of LRM varies widely with the technology available, the mix of residential and commercial equipment, and the skills of the servicing workforce, so the strategy carries low confidence.

Most countries regulate the handling and collection of fluorinated gases under their obligations to the Montreal Protocol on Substances that Deplete the Ozone Layer. Norway's tax-and-refund scheme and Australia's Refrigerant Reclaim programme are examples of national end-of-life collection systems. Many countries also run technician training programs, often supported by UNEP OzonAction, to reduce leaks during servicing (UNEP OzonAction, n.d.).

5.11 other-mining-quarrying - Electrification of mine equipment

Greenhouse gas emissions from the mineral extraction sector are primarily generated by the combustion of fuels from mobile sources, such as vehicles and machinery, and fixed sources like generators, particularly at sites without access to electricity grids. Emissions can be significantly reduced by replacing diesel-powered mining vehicles and plant machinery with electric equivalents, and connecting sites to local grids to avoid onsite electricity generation. This approach has the potential to fully eliminate direct emissions at some mining sites where entire vehicle fleets can be electrified. Electrification is increasingly adopted across the sector, as documented by

numerous mining companies. However, practical constraints—including the size of operations, the number of vehicles, charging capacity, costs, and limitations on vehicle range—may restrict the feasibility of complete adoption (Schmidt et al., 2025).

5.12 textiles-leather-apparel - Energy efficiency + heat pumps

The textile industry accounts for about 2% of global greenhouse gas emissions (Hasanbeigi and Zuberi, 2022). Fiber and yarn production, together with fabric dyeing and finishing, account for around 90% of the sector's emissions, largely from fossil fuels burned to raise steam and hot water. This strategy combines energy-efficiency measures with industrial heat pumps, which supply process heat with electricity at a much lower energy intensity than fossil-fired boilers. In wet processing, heat pumps can also recover heat from wastewater; integrating them with an optimized heat exchanger network has been shown to cut total annualized costs by up to 43% (Kim et al., 2022).

Industrial heat pumps can reduce energy consumption by 10–30%, depending on the manufacturing process, as shown in Table 9. The strategy is applied by facility type. Wet processing, the most heat-intensive stage, takes the largest reduction. Each variant applies to all activity at the asset, with medium confidence. The additional electricity demand is accounted for in the electricity-generation sector through the induced activity conversion rate.

Table 9 Facility types with heat pumps strategy applied and the resulting CO_2 emissions ratio with induced electricity conversion rate.

Facility type	Measures	CO_2 emissions factor ratio (new ef/old ef)	Induced electricity conversion rate
Wet processing	Heat pumps, heat recovery, process optimization	0.70	3.29
Textile production	Heat pumps, efficient motors, waste heat recovery, process optimization	0.90	1.49
Garment and product manufacturing	Heat pumps, efficient equipment, facility energy management	0.90	0.60
General (type unknown)	Heat pumps, process electrification, waste heat recovery, advanced energy management	0.85	2.40

Industrial heat pumps are being trialed in textile manufacturing, with feasibility assessments in India, Vietnam and China, grant-funded steam heat pump installations in Vietnam and China, and hot-water heat pump pilots in Italy and Türkiye (Aii, 2026). Current limitations include temperature (commercially mature steam heat pumps reach about 120°C, below the 180°C+ needed for heat-setting), unit size (a single steam heat pump rarely exceeds 2 MW, against boilers averaging about 17.6 MW at surveyed facilities) and cost, with typical payback of five to eight years. Near-term electrification is therefore likely to be partial.

5.13 wood-and-wood-products - Energy from biomass

The production of wood and wood products generates greenhouse gas emissions primarily from fuel combustion in mill equipment and rolling stock, accounting for approximately 0.03% of global CO2-equivalent emissions. A practical approach to reducing these emissions is the use of excess forest residuals as biomass to replace fossil fuels in the production process. By utilizing these residuals for heat energy in mills, net emissions can be reduced by up to 66%. Some mills have already implemented the use of forest residues for power or heating, demonstrating feasibility, though wider adoption may be influenced by financial and technological constraints globally (Brandeis, 2024; FPJ, 2024).

5.14 enteric-fermentation-other - Feed additives in ruminant diet

Feed additives in ruminant diets offer a strategy to reduce methane emissions from enteric fermentation in non-cattle livestock, including sheep, goats, and yaks, which together account for approximately 1.2% of global greenhouse gas emissions. These additives are methane-inhibiting, suppressing the bacteria responsible for methane production in the digestive system, and thereby lowering emissions. Using compounds, such as 3-NOP, have shown emissions reductions of around 20% in enteric fermentation methane emissions.

Feed additives adoption has been implemented in various countries, including the EU, the U.S.A., and Brazil, primarily to enhance beef cattle weight gain or milk production, with reductions in methane emissions considered a secondary benefit. Despite this, global uptake remains limited due to regulatory restrictions in some regions, costs associated with implementation, potential impacts on product quality and prices, farmers' unfamiliarity with the technology, and consumer skepticism.

5.15 cropland-fires - Residue removal without burning

Crop residues are the leftover crop biomass remaining after harvest, such as straw or stover. Rather than burning these residues, the recommended approach is to remove them without combustion, which eliminates emissions associated with burning. This practice avoids the release of carbon dioxide, methane, and nitrous oxide that would otherwise result from the combustion of leftover crop residues.

In many agricultural contexts, crop residues have additional value as animal feed, bedding material, or organic soil amendments. In regions with diversified agricultural economies, established markets for these byproducts create economic incentives for adopting residue removal rather than burning. However, some farmers continue to burn crop residues due to time constraints associated with double cropping, as they need to clear fields quickly to plant the next crop. More mechanized agricultural systems are better equipped to manage residues efficiently, with the capacity to collect and transport them for alternative uses. The effectiveness of this approach therefore depends on the degree of mechanization, market access for residue byproducts, and timing pressures within local cropping systems.

Supplementary material

Table S1 Climate TRACE, CEDS, EDGAR, and FAOSTAT Sector Names.

Data Source	**Original Inventory Sector Name**
FAOSTAT	Enteric Fermentation
FAOSTAT	Manure Management
FAOSTAT	Manure applied to Soils
FAOSTAT	Manure left on Pasture
FAOSTAT	Synthetic Fertilizers
FAOSTAT	Rice Cultivation
FAOSTAT	Burning - Crop residues
FAOSTAT	Crop Residues
FAOSTAT	On-farm energy use
FAOSTAT	Savanna fires
FAOSTAT	Fires in humid tropical forests
FAOSTAT	Forest fires
FAOSTAT	Fires in organic soils
FAOSTAT	Drained organic soils
FAOSTAT	Drained organic soils
FAOSTAT	Net Forest conversion
FAOSTAT	Forestland
EDGAR	Main Activity Electricity and Heat Production
EDGAR	Petroleum Refining - Manufacture of Solid Fuels and Other Energy Industries
EDGAR	Manufacturing Industries and Construction
EDGAR	Civil Aviation

Data Source	Original Inventory Sector Name
EDGAR	Road Transportation no resuspension
EDGAR	Railways
EDGAR	Water-borne Navigation
EDGAR	Other Transportation
EDGAR	Other Sectors
EDGAR	Non-Specified
EDGAR	Solid Fuels
EDGAR	Oil and Natural Gas
EDGAR	Cement production
EDGAR	Lime production
EDGAR	Glass Production
EDGAR	Other Process Uses of Carbonates
EDGAR	Chemical Industry
EDGAR	Metal Industry
EDGAR	Non-Energy Products from Fuels and Solvent Use
EDGAR	Liming
EDGAR	Urea application
EDGAR	Other
EDGAR	Enteric Fermentation
EDGAR	Manure Management
EDGAR	Emissions from biomass burning
EDGAR	Rice cultivations
EDGAR	Solid Waste Disposal
EDGAR	Biological Treatment of Solid Waste
EDGAR	Incineration and Open Burning of Waste
EDGAR	Wastewater Treatment and Discharge
EDGAR	Other
EDGAR	Direct N2O Emissions from managed soils
EDGAR	Indirect N2O Emissions from managed soils

Data Source	Original Inventory Sector Name
EDGAR	Indirect N2O Emissions from manure management
EDGAR	Indirect N2O emissions from the atmospheric deposition of nitrogen in NOx and NH3
EDGAR	Other
CEDS	1A1a_Electricity-autoproducer
CEDS	1A1a_Electricity-public
CEDS	1A1a_Heat-production
CEDS	1A1bc_Other-transformation
CEDS	, 1A5_Other-unspecified
CEDS	1B1_Fugitive-solid-fuels
CEDS	1B2_Fugitive-petr
CEDS	1B2b_Fugitive-NG-prod
CEDS	1B2d_Fugitive-other-energy
CEDS	1B2b_Fugitive-NG-distr
CEDS	2A1_Cement-production
CEDS	1A2a_Ind-Comb-Iron-steel
CEDS	2C1_Iron-steel-alloy-prod
CEDS	1A2b_Ind-Comb-Non-ferrous-metals,
CEDS	2C3_Aluminum-production
CEDS	2C4_Non-Ferrous-other-metals
CEDS	1A2c_Ind-Comb-Chemicals
CEDS	2B_Chemical-industry
CEDS	2B2_Chemicals-Nitric-acid
CEDS	2B3_Chemicals-Adipic-acid
CEDS	1A2d_Ind-Comb-Pulp-paper
CEDS	1A2g_Ind-Comb-mining-quarying
CEDS	1A2e_Ind-Comb-Food-tobacco
CEDS	1A2g_Ind-Comb-textile-leather
CEDS	1A2g_Ind-Comb-wood-products
CEDS	2A2_Lime-production
CEDS	2Ax_Other-minerals

Data Source	Original Inventory Sector Name
CEDS	1A2f_Ind-Comb-Non-metalic-minerals
CEDS	1A2g_Ind-Comb-Construction
CEDS	1A2g_Ind-Comb-machinery
CEDS	1A2g_Ind-Comb-other
CEDS	1A2g_Ind-Comb-transpequip
CEDS	2D_Chemical-products-manufacture-processing,
CEDS	2D_Other-product-use
CEDS	2D_Degreasing-Cleaning,
CEDS	2D_Paint-application,
CEDS	2H_Pulp-and-paper-food-beverage-wood
CEDS	1A3aii_Domestic-aviation
CEDS	1A3ai_International-aviation
CEDS	1A3dii_Domestic-navigation
CEDS	1A3di_International-shipping
CEDS	1A3di_Oil_Tanker_Loading
CEDS	1A3b_Road
CEDS	1A3c_Rail
CEDS	1A3eii_Other-transp
CEDS	1A4b_Residential
CEDS	1A4a_Commercial-institutional
CEDS	1A4c_Agriculture-forestry-fishing
CEDS	3E_Enteric-fermentation
CEDS	3B_Manure-management
CEDS	3D_Rice-Cultivation
CEDS	3D_Soil-emissions
CEDS	3I_Agriculture-other
CEDS	7BC_Indirect-N2O-non-agricultural-N
CEDS	5A_Solid-waste-disposal
CEDS	5E_Other-waste-handling
CEDS	5C_Waste-combustion

Data Source	Original Inventory Sector Name
CEDS	5D_Wastewater-handling
ClimateTRACE	electricity-generation
ClimateTRACE	heat-plants
ClimateTRACE	other-energy-use
ClimateTRACE	domestic-aviation
ClimateTRACE	international-aviation
ClimateTRACE	road-transportation
ClimateTRACE	railways
ClimateTRACE	international-shipping
ClimateTRACE	domestic-shipping
ClimateTRACE	other-transport
ClimateTRACE	residential-onsite-fuel-usage
ClimateTRACE	non-residential-onsite-fuel-usage
ClimateTRACE	other-onsite-fuel-usage
ClimateTRACE	coal-mining
ClimateTRACE	other-solid-fuels
ClimateTRACE	oil-and-gas-production
ClimateTRACE	oil-and-gas-transport
ClimateTRACE	oil-and-gas-refining
ClimateTRACE	cement
ClimateTRACE	chemicals
ClimateTRACE	petrochemicals-steam-cracking
ClimateTRACE	other-chemicals
ClimateTRACE	iron-and-steel
ClimateTRACE	aluminum
ClimateTRACE	other-metals
ClimateTRACE	pulp-and-paper
ClimateTRACE	glass
ClimateTRACE	lime
ClimateTRACE	food-beverage-tobacco

Data Source	Original Inventory Sector Name
ClimateTRACE	wood-and-wood-products
ClimateTRACE	textiles-leather-apparel
ClimateTRACE	other-manufacturing
ClimateTRACE	bauxite-mining
ClimateTRACE	copper-mining
ClimateTRACE	iron-mining
ClimateTRACE	rock-quarrying
ClimateTRACE	sand-quarrying
ClimateTRACE	other-mining-and-quarrying
ClimateTRACE	enteric-fermentation-cattle-operation
ClimateTRACE	enteric-fermentation-cattle-pasture
ClimateTRACE	enteric-fermentation-other
ClimateTRACE	manure-management-cattle-operation
ClimateTRACE	manure-left-on-pasture-cattle
ClimateTRACE	manure-management-other
ClimateTRACE	rice-cultivation
ClimateTRACE	synthetic-fertilizer-application
ClimateTRACE	crop-residues
ClimateTRACE	manure-applied-to-soils
ClimateTRACE	cropland-fires
ClimateTRACE	other-agricultural-soil-emissions
ClimateTRACE	solid-waste-disposal
ClimateTRACE	biological-treatment-of-solid-waste-and-biogenic
ClimateTRACE	incineration-and-open-burning-of-waste
ClimateTRACE	industrial-wastewater-treatment-and-discharge
ClimateTRACE	domestic-wastewater-treatment-and-discharge
ClimateTRACE	fluorinated-gases
ClimateTRACE	forest-land-fires
ClimateTRACE	forest-land-clearing
ClimateTRACE	forest-land-degradation

Data Source	Original Inventory Sector Name
ClimateTRACE	net-forest-land
ClimateTRACE	net-shrubgrass
ClimateTRACE	net-wetland
ClimateTRACE	shrub grass-fires
ClimateTRACE	wetland-fires
ClimateTRACE	removals
ClimateTRACE	water-reservoirs

Table S2 The crosswalk table: Climate TRACE sector and related source data

Sector on Climate TRACE site	Equation to calculate	Gas	Filled data source
Other Energy Use	= EDGAR[“Main Activity Electricity and Heat Production”] - ClimateTrace[“electricity-generation”] - ClimateTrace[“heat-plants”]	CO2 CH4 N2O	EDGAR
Railways	= EDGAR[“Railways”]	CO2 CH4 N2O	EDGAR
Other Transport	= EDGAR[“Other Transportation”]	CO2 CH4 N2O	EDGAR
Other Onsite Fuel Usage	= EDGAR[“Non-Specified”]	CO2 CH4 N2O	EDGAR
Other Solid Fuels	= EDGAR[“Solid Fuels”] - ClimateTrace[“coal-mining”]	CO2 CH4 N2O	EDGAR
Other Fossil Fuel Operations	= EDGAR[“Oil and Natural Gas”] + EDGAR[“Petroleum Refining - Manufacture of Solid Fuels and Other Energy Industries”] - ClimateTrace[“oil-and-gas-production”] - ClimateTrace[“oil-and-gas-transport”] - ClimateTrace[“oil-and-gas-refining”]	CO2 CH4 N2O	EDGAR
Other Manufacturing	= CEDS[“Construction Combustion”] + CEDS[“Machinery Combustion”] + CEDS[“Transportation Equipment Combustion”] +	CO2 CH4 N2O	CEDS EDGAR

Sector on Climate TRACE site	Equation to calculate	Gas	Filled data source
	CEDS[“Other Combustion”] + CEDS[“Other Minerals”] + CEDS[“Chemical Products Manufacturing and Processing”] + CEDS[“Degreasing and Cleaning”] + CEDS[“Other Product Use”] + CEDS[“Paint Application”] + “Misc. Mineral Industry Combustion” (see Section 2.2.2.2) - EDGAR[“Glass Production”]		
Enteric Fermentation Other	=FAOSTAT[“enteric-fermentation- non cattle”]	CH4	FAOSTAT
Manure Management Other	=FAOSTAT[“manure-management- non cattle”]	CH4	FAOSTAT
Other Agricultural Soil Emissions	=FAOSTAT[“manure-left-on-pasture - noncattle”] + FAOSTAT[“drained-organic-soils”]+ EDGAR[“Liming”] + EDGAR[“Urea application”]	CO2 N2O	FAOSTAT
Other Agricultural Soil Emissions	=FAOSTAT[“crop residues”] + EDGAR[“Liming”] + EDGAR[“Urea application”]	CH4	FAOSTAT
Biological Treatment of Solid Waste and Biogenic	= EDGAR[“Biological Treatment of Solid Waste”]	CO2 CH4 N2O	EDGAR
Incineration and Open Burning of Waste	= EDGAR[“Incineration and Open Burning of Waste”]	CO2 CH4 N2O	EDGAR
Fluorinated Gases	= EDGAR[fluorinated-gases]	f-gases	EDGAR
Other Mining and Quarrying	= CEDS[“mining & quarrying combustion"] - ClimateTRACE[“copper-mining”] - ClimateTRACE[“bauxite-mining”] - ClimateTRACE[“iron-mining”] - ClimateTRACE[“rock-quarrying”] - ClimateTRACE[“sand-quarrying”]	CO2 CH4 N2O	CEDS
Other Chemicals	= EDGAR[“Chemical Industry”] + CEDS[“Chemical Industry Combustion”] - ClimateTRACE[“chemicals”] - ClimateTRACE[“petrochemicals-steam-	CO2 CH4 N2O	EDGAR CEDS

Sector on Climate TRACE site	Equation to calculate	Gas	Filled data source
	cracking”]		
Wood and Wood Products	= CEDS[“Wood, wood products combustion”]	CO2 CH4 N2O	CEDS
Other Metals	= EDGAR[“Metal Industry”] + CEDS[“Non-Ferrous metals combustion”] - ClimateTRACE[“iron-and-steel”] - ClimateTRACE[“aluminum”]	CO2 CH4 N2O	EDGAR CEDS

Table S3 Climate TRACE sectors with IPCC equivalent sector coverage. Sector specific global emission (V5.11.0) contributions to the 2025 global total excluding LULUCF sectors, estimation method (from Introduction), data sources to generate emission estimates are provided.

Sector	Subsector	IPCC Category	% of Global Emissions w/o LULUCF (Based on CT inventory)	Estimation Method	Data Source
power	electricity-generation	1.A.1.a.i Electricity Generation 1.A.1.a.ii Combined Heat and Power Generation	23.44	1	WattTime, Transition Zero, Global Energy Monitor
power	heat-plants	1.A.1.a.iii Heat plants	0.10%	3	WattTime
power	other-energy-use	1.A.1.a.iv Other	3.44%	5	EDGAR
buildings	residential-onsite-fuel-usage	1.A.4.b Residential Stationary Combustion	4.84%	2	Duke University
buildings	non-residential-onsite-fuel-usage	1.A.4.a Commercial/institutional Stationary Combustion 1.A.4.c Agriculture/Forestry/Fishing	1.24%	2	Duke University
buildings	other-onsite-fuel-usage	1.A.5 Non-Specified	0.29%	5	EDGAR
manufacturing	petrochemical-steam-cracking	2.B.8.b Ethylene	0.51%	3	RMI
manufacturing	iron-and-steel	1.A.2.a Iron and Steel 2.C.1 Iron and Steel Production	4.58%	1	Transition Zero
manufacturing	cement	1.A.2.f Non-metallic Minerals 2.A.1 Cement Production	4.17%	1	Transition Zero
manufacturing	chemicals	1.A.2.c Chemicals 2.B.1 Ammonia Production 2.B.7 Soda Ash Production 2.B.8.a Methanol	1.07%	3	Transition Zero
manufacturing	aluminum	1.A.2.b Non-Ferrous Metals 2.C.3 Aluminum Production	0.56%	3	Transition Zero
manufacturing	pulp-and-paper	1.A.2.d Pulp, Paper and Print	0.03%	3	Transition Zero

Sector	Subsector	IPCC Category	% of Global Emissions w/o LULUCF (Based on CT inventory)	Estimation Method	Data Source
manufacturing	glass	1.A.2.f Non-metallic Minerals 2.A.3 Glass production	0.05%	4	WattTime
manufacturing	lime	1.A.2.f Non-metallic Minerals 2.A.2 Lime Production	0.89%	4	WattTime
manufacturing	food-beverage-tobacco	1.A.2.e Food Processing, Beverages and Tobacco 2.H.2 Food and Beverages Industry	0.44%	4	WattTime
manufacturing	wood-and-wood-products	1.A.2.g.iv Wood and Wood Products	0.02%	5	WattTime
manufacturing	textiles-leather-apparel	1.A.2.g.vi Textile and Leather	0.08%	4	WattTime
manufacturing	other-metals	1.A.2.b Non-Ferrous Metals 2.C.2 Ferroalloys Production 2.C.4 Magnesium Production 2.C.5 Lead Production 2.C.6 Zinc Production 2.C.7 Other	0.14%	3,5	WattTime
manufacturing	other-chemicals	1.A.2.c Chemicals 2.B.2 Nitric Acid Production 2.B.3 Adipic Acid Production 2.B.4 Caprolactam, Glyoxal and Glyoxylic Acid Production 2.B.5 Carbide Production 2.B.6 Titanium Dioxide Production 2.B.8.c Ethylene Dichloride and Vinyl Chloride Monomer 2.B.8.d Ethylene Oxide 2.B.8.e Acrylonitrile 2.B.8.f Carbon Black 2.B.8.g Other	1.34%	3,5	WattTime
manufacturing	other-manufacturing	1.A.2.g.i Manufacturing of Machinery 1.A.2.g.ii Manufacturing of Transport Equipment 1.A.2.g.v Construction 1.A.2.g.vii Off-road Vehicles and Other Machinery 1.A.2.g.viii Other 2.A.4 Other Process Uses of Carbonates 2.D Non-energy Products from Fuels and Solvent Use 2.G Other Product Manufacture and Use	3.23%	3,5	EDGAR
fossil-fuel-operations	oil-and-gas-refining	1.A.1.b Petroleum Refining	1.67%	3	RMI
fossil-fuel-operations	oil-and-gas-production	1.A.1.c.ii Oil and Gas Extraction 1.B.2.a.i Exploration 1.B.2.a.ii Production 1.B.2.b.i Exploration	5.72%	2	RMI

Sector	Subsector	IPCC Category	% of Global Emissions w/o LULUCF (Based on CT inventory)	Estimation Method	Data Source
		1.B.2.b.ii Production 1.B.2.b.iii Processing 1.B.2.c Venting and Flaring			
fossil-fuel-operations	oil-and-gas-transport	1.B.2.a.iii Transport 1.B.2.b.iv Transmission and Storage	1.37%	2	RMI
fossil-fuel-operations	other-solid-fuels	1.B.1.b Solid Fuel Transformation	1.32%	5	EDGAR
fossil-fuel-operations	coal-mining	1.B.1.a Coal Mining and Handling	4.01%	3	WattTime, Global Energy Monitor
mineral-extraction	copper-mining	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.15%	3	Hypervine
mineral-extraction	iron-mining	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.11%	3	Hypervine
mineral-extraction	bauxite-mining	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.01%	3	Hypervine
mineral-extraction	sand-quarrying	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.001%	3	Hypervine
mineral-extraction	rock-quarrying	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.002%	3	Hypervine
mineral-extraction	other-mining-quarrying	1.A.2.g.iii Mining (Excluding Fuels) and Quarrying	0.07%	5	CEDS
transportation	domestic-aviation	1.A.3.a Domestic Aviation	0.54%	3	WattTime
transportation	international-aviation	International Aviation	1.01%	3	WattTime
transportation	road-transportation	1.A.3.b Road Transportation	11.44%	1	Johns Hopkins University Applied Physics Laboratory
transportation	domestic-shipping	1.A.3.d. Domestic Navigation	0.93%	2	OceanMind, Global Fishing Watch, UCSB emLab
transportation	international-shipping	International Navigation	0.96%	2	OceanMind, Global Fishing Watch, UCSB emLab
transportation	non-broadcasting-vessels	N/a	0.70%	1	Global Fishing Watch, UCSB emLab
transportation	railways	1.A.3.c Railways	0.17%	5	EDGAR
transportation	other-transport	1.A.3.e Other Transportation	0.32%	5	EDGAR
agriculture	rice-cultivation	3.C Rice Cultivation	1.32%	2	University of Malaysia Terengganu, Carbon Yield
agriculture	cropland-fires	3.F Field Burning of Agricultural Residues	0.08%	2	Michigan State University, Carbon Yield
agriculture	enteric-fermentation-cattle-operation	3.A.1 Cattle	1.51%	2	WattTime, Saint Louis

Sector	Subsector	IPCC Category	% of Global Emissions w/o LULUCF (Based on CT inventory)	Estimation Method	Data Source
					University, Earth Genome
agriculture	enteric-fermentation-cattle-pasture	3.A.1 Cattle	0.99%	2	Saint Louis University
agriculture	enteric-fermentation-other	3.A.2 Sheep, 3.A.3 Swine, 3.A.4 Other	1.34%	5	WattTime analysis of FAOSTAT data
agriculture	manure-management-cattle-operation	3.B.1 Cattle	0.40%	2	WattTime, Saint Louis University, Earth Genome
agriculture	manure-left-on-pasture-cattle	3.D.1.c. Urine and Dung Deposited by Grazing Animals	0.31%	2	Saint Louis University
agriculture	manure-management-other	3.B.2 Sheep, 3.B.3 Swine, 3.B.3 Other	0.92%	5	WattTime analysis of FAOSTAT data
agriculture	synthetic-fertilizer-application	3.D.1.a Inorganic N Fertilizers, 3.D.2 Indirect N_2O Emissions From Managed Soils	0.93%	3	Michigan State University
agriculture	crop-residues	3.D.1.d Crop Residues	0.21%	5	FAOSTAT
agriculture	manure-applied-to-soils	3.D.1.b.i Animal Manure Applied to Soils	0.25%	5	FAOSTAT
agriculture	other-agricultural-soil-emissions	3.D.1.f Cultivation of Organic Soils 3.G Liming 3.H Urea Application	1.80%	5	FAOSTAT and EDGAR
waste	solid-waste-disposal	5.A Solid Waste Disposal	1.83%	2 (source) 5 (country)	RMI
waste	biological-treatment-of-solid-waste-&-biogenic	5.B Biological Treatment of Solid Waste	0.04%	5	EDGAR
waste	incineration-and-open-burning-of-waste	5.C Incineration and Open Burning of Waste	0.03%	5	EDGAR
waste	domestic-wastewater-treatment-and-discharge	5.D.1 Domestic Wastewater Treatment and Discharge	1.34%	2	Johns Hopkins Applied Physics Laboratory
waste	industrial-wastewater-treatment-and-discharge	5.D.2 Industrial Wastewater Treatment and Discharge	0.28%	2	Johns Hopkins Applied Physics Laboratory
fluorinated-gases	fluorinated-gases	2.B Chemical Industry (HFCs and PFCs) 2.C Metal Industry (HFCs and PFCs) 2.E Electronics Industry (HFCs and PFCs) 2.F Product Uses as Substitutes for Ozone Depleting Substances (HFCs and PFCs) 2.G Other Product Manufacture and Use (HFCs and PFCs)	2.93%	5	EDGAR
forestry-and-land-use	water-reservoirs	4.D.1.b Flooded Land Remaining	0.17%	2	Johns Hopkins

Sector	Subsector	IPCC Category	% of Global Emissions w/o LULUCF (Based on CT inventory)	Estimation Method	Data Source
		Flooded Land			Applied Physics Laboratory
forestry-and-land-use	net-forest	4.A Forest Land	-1.17%	1	CTrees
forestry-and-land-use	net-shrubgrass	4.C Grassland	-2.60%	1	CTrees
forestry-and-land-use	net-wetland	4.D.1.c Other Wetlands Remaining Other Wetlands, 4.D.2. Land Converted to Wetlands	-0.43%	1	CTrees

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Geographic boundaries and names (iso3_country data attribute): The depiction and use of boundaries, geographic names and related data shown on maps and included in lists, tables, documents, and databases on Climate TRACE are generated from the Global Administrative Areas (GADM) project (Version 4.1 released on 16 July 2022) along with their corresponding ISO3 codes, and with the following adaptations:

- HKG (China, Hong Kong Special Administrative Region) and MAC (China, Macao Special Administrative Region) are reported at GADM level 0 (country/national);
- Kosovo has been assigned the ISO3 code ‘XKX’;
- XCA (Caspian Sea) has been removed from GADM level 0 and the area assigned to countries based on the extent of their territorial waters;
- XAD (Akrotiri and Dhekelia), XCL (Clipperton Island), XPI (Paracel Islands) and XSP (Spratly Islands) are not included in the Climate TRACE dataset;
- ZNC name changed to ‘Turkish Republic of Northern Cyprus’ at GADM level 0;

- The borders between India, Pakistan and China have been assigned to these countries based on GADM codes Z01 to Z09.
- Two IDs have been created for a region in UKR with missing IDs (at Level 1 and Level 2).
- UNK added to GADM level 0 to denote 'unknown' countries, which primarily applies to non-broadcasting-vessels whose port berthing locations are not known.
- TUR name changed to “Türkiye”
- SWZ name changed to “Eswatini”
- Missing Con Dao Island added as VNM.7.X_1 and Kili Island added as MHL.X_1.

The above usage is not warranted to be error free and does not imply the expression of any opinion whatsoever on the part of Climate TRACE Coalition and its partners concerning the legal status of any country, area or territory or of its authorities, or concerning the delimitation of its borders.

Disclaimer: The emissions provided for this sector are our current best estimates of emissions, and we are committed to continually increasing the accuracy of the models on all levels. Please review our terms of use and the sector-specific methodology documentation before using the data. If you identify an error or would like to participate in our data validation process, please contact us.

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